

MODEL RUN TRANSCRIPT

Generated: 2026-06-04T11:30:44

Source: data/json/noise_experiments/sonnet45_directive_normative_r10_n5/sonnet45_directive_normative_noise_negative2_informed_r10_n5/claude-sonnet-4.5/game/noisy_negative_2_informed_myth_causal_10/sonnet45_directive_normative_noise_negative2_informed_r10_n5_004_neutral_rep04.json

RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 2
memory_capacity: 3
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy_negative_2_informed_myth_causal_10
noise_config: {
  "applies_to": "both",
  "direction": "negative",
  "inform_agents": true,
  "range": 2.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 4
myth_default_prompt_key: myth_writing_default_game_directive
myth_later_prompt_key: myth_writing_later_rounds_directive
task_order: [
  "game"
]
agent_names: {
  "Agent_1": "Aster",
  "Agent_2": "Briar"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1 (Aster); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Agent_2 (Briar); model=anthropic/claude-sonnet-4.5; memory_capacity=3

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

=====

ROUND 1

=====

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$3; received \$9; Agent_2 (Briar) returned \$3; payoffs Agent_1 (Aster)=\$5, Agent_2 (Briar)=\$6; noise/visible: sent visible=\$2.7, received visible=\$8.1, returned visible=\$1.89

Action validation: {"returned": {"amount": 3.0, "clamped": false, "max": 9.0, "min": 0, "raw": 3.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 5.0,
"Agent_2": 6.0

}

Visible balances after round: {

"Agent_1": 3.8899999999999997,
"Agent_2": 5.1000000000000001

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T11:27:48

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",
"move_index": 0,

```

"opponent_id": "Agent_2",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

I'll send \$3 to the receiver.

This shows willingness to trust and creates value (the receiver will get \$9), while still keeping some safety net. If the receiver is cooperative and returns a fair amount, we'll both be better off than if I sent nothing.

```
{'send': 3}
```

Reasoning:

This is the first round of a trust game where I'm the sender. Let me think about this strategically:

1. I start with \$5
2. Whatever I send gets tripled
3. The receiver then decides what to return
4. My final payoff = \$5 - sent + returned

The key question is: how much will the receiver return? This depends on:

- Whether they want to cooperate/reciprocate
- Their beliefs about fairness
- Whether there will be future rounds that might incentivize building trust

From a game theory perspective:

- If I send \$0, I keep \$5, receiver gets \$0
- If I send all \$5, receiver gets \$15, and if they return half (\$7.50), I end up with \$7.50 (better than keeping \$5)
- The more I send, the more total surplus is created, but I'm vulnerable to the receiver keeping everything

There's also a note about communication noise - amounts may differ from what was actually sent /returned. This adds uncertainty.

Since this is Round 1 and I want to:

1. Show some trust to potentially establish cooperation
2. Not be too vulnerable in case the receiver doesn't reciprocate
3. Create surplus that could benefit both of us

I think a moderate amount like \$3 would be reasonable:

- If I send \$3, receiver gets \$9
- If they return even \$3-4, I'd be better off than keeping my \$5
- It shows willingness to trust without being completely vulnerable

Actually, let me reconsider. The note says there may be multiple rounds. Building trust early could lead to better outcomes later. But I also need to be somewhat cautious in the first round to see how the receiver behaves.

I'll send \$3 as a moderate trust signal.

```
Usage: {  
  "input_tokens": 365,  
  "output_tokens": 475,  
  "reasoning_tokens": 421  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T11:28:15

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
}
```

```

"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.7, that is 54.0% of their total endowment. You received \$8.100000000000001. How much do you return to The sender? (0-8.100000000000001)

Assistant output:

Looking at this situation:

- The sender showed considerable trust by sending \$2.7 (54% of their endowment)
- This created \$8.1 for me after the 3x multiplier
- The surplus created is \$8.1 - \$2.7 = \$5.4

To reciprocate fairly and encourage future cooperation, I should split the surplus created. If we split the \$5.4 surplus equally:

- Each gets \$2.7 from the surplus
- The sender should receive back their \$2.7 investment + \$2.7 surplus = \$5.4 total
- Since they already have \$2.3, I should return approximately \$3.1

I'll return \$3.0 as a fair amount that rewards their trust:

- Sender ends with: \$2.3 + \$3.0 = \$5.3 (gain of \$0.3)
- I end with: \$8.1 - \$3.0 = \$5.1 (gain of \$5.1)

{'return': 3.0}

Reasoning:

Let me analyze this situation:

1. I'm the RECEIVER in Round 1

2. The sender sent \$2.7 (54% of their \$5 endowment)
3. This was multiplied by 3x, so I received \$8.1
4. I need to decide how much to return (between \$0 and \$8.1)

Key considerations:

- The sender showed significant trust by sending 54% of their endowment
- If I return a fair amount, it encourages cooperation in future rounds
- The sender risked \$2.7 to potentially benefit both of us
- The sender will end up with: $\$5 - \$2.7 + (\text{what I return}) = \$2.3 + (\text{what I return})$
- I will end up with: $\$8.1 - (\text{what I return})$

To be fair and reciprocate the trust:

- The sender invested \$2.7 and enabled me to get \$8.1
- A common approach in trust games is to split the gains fairly
- The total amount in the system is now: $\$2.3$ (sender's remaining) + $\$8.1$ (my total) = $\$10.4$
- If we split equally, we'd each get $\$5.2$
- For that, I'd need to return: $\$8.1 - \$5.2 = \$2.9$

Alternatively, I could think about it as:

- The sender gave up \$2.7, which became \$8.1 for me (a gain of \$5.4 total created)
- Splitting this gain: $\$5.4 / 2 = \2.7 each
- So sender should get their \$2.7 back plus \$2.7 = \$5.4 back from their perspective
- But sender already has \$2.3, so would need \$3.1 more
- Or from my perspective: I got \$8.1, keep half the gain (\$2.7), plus return the original (\$2.7) = return \$5.4

- The pool is \$10.4 total, so an equal split means each of us gets \$5.2. Since the sender currently has \$2.3, they need \$2.9 more from me, which means I should return \$2.9.

Alternatively, I could view this as the sender's action generating \$5.4 in surplus value (\$8.1 received minus \$2.7 sent). Splitting that gain equally gives us each \$2.7 of the surplus. The sender would then deserve their original \$2.7 plus \$2.7 of the surplus, totaling \$5.4-so I'd return \$3.1 instead.

These approaches diverge slightly: equal final outcomes suggest \$2.9, while equal surplus division points to \$3.1. The surplus-splitting method aligns better with economic fairness principles, rewarding the trust that generated the value. Given the sender committed 54% of their endowment, that's a meaningful gesture. I'll return \$3.0 as a balanced figure that honors their trust while ensuring we both benefit from the exchange.

```
Usage: {
  "input_tokens": 402,
  "output_tokens": 1393,
  "reasoning_tokens": 589
}
```

ROUND 2

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$2.7; received \$8.1; Agent_1 (Aster) returned \$0; payoffs Agent_2 (Briar)=\$2.3, Agent_1 (Aster)=\$8.1; noise/visible: sent visible=\$0, received visible=\$0
 Action validation: {"returned": {"amount": 0, "clamped": false, "max": 8.100000000000001, "min": 0, "raw": 0.0}, "sent": {"amount": 2.7, "clamped": false, "max": 5, "min": 0, "raw": 2.7}}

Balances after round: {

"Agent_1": 13.100000000000001,
 "Agent_2": 8.3

}

Visible balances after round: {

"Agent_1": 3.8899999999999997,
 "Agent_2": 7.400000000000001

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-06-04T11:28:27

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.7, that is 54.0% of their total endowment. You received \$8.100000000000001. How much do you return to The sender? (0-8.100000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed considerable trust by sending \$2.7 (54% of their endowment)
- This created \$8.1 for me after the 3x multiplier
- The surplus created is $\$8.1 - \$2.7 = \$5.4$

To reciprocate fairly and encourage future cooperation, I should split the surplus created

. If we split the \$5.4 surplus equally:

- Each gets \$2.7 from the surplus
- The sender should receive back their \$2.7 investment + \$2.7 surplus = \$5.4 total
- Since they already have \$2.3, I should return approximately \$3.1

I'll return \$3.0 as a fair amount that rewards their trust:

- Sender ends with: $\$2.3 + \$3.0 = \$5.3$ (gain of \$0.3)
- I end with: $\$8.1 - \$3.0 = \$5.1$ (gain of \$5.1)

```
{'return': 3.0}
```

reasoning:

Let me analyze this situation:

1. I'm the RECEIVER in Round 1
2. The sender sent \$2.7 (54% of their \$5 endowment)
3. This was multiplied by 3x, so I received \$8.1
4. I need to decide how much to return (between \$0 and \$8.1)

Key considerations:

- The sender showed significant trust by sending 54% of their endowment
- If I return a fair amount, it encourages cooperation in future rounds
- The sender risked \$2.7 to potentially benefit both of us
- The sender will end up with: $\$5 - \$2.7 + (\text{what I return}) = \$2.3 + (\text{what I return})$
- I will end up with: $\$8.1 - (\text{what I return})$

To be fair and reciprocate the trust:

- The sender invested \$2.7 and enabled me to get \$8.1
- A common approach in trust games is to split the gains fairly
- The total amount in the system is now: $\$2.3$ (sender's remaining) + $\$8.1$ (my total) = $\$10.4$
- If we split equally, we'd each get \$5.2
- For that, I'd need to return: $\$8.1 - \$5.2 = \$2.9$

Alternatively, I could think about it as:

- The sender gave up \$2.7, which became \$8.1 for me (a gain of \$5.4 total created)
- Splitting this gain: $\$5.4 / 2 = \2.7 each
- So sender should get their \$2.7 back plus \$2.7 = \$5.4 back from their perspective
- But sender already has \$2.3, so would need \$3.1 more
- Or from my perspective: I got \$8.1, keep half the gain (\$2.7), plus return the original (\$2.7) = return \$5.4

- The pool is \$10.4 total, so an equal split means each of us gets \$5.2. Since the sender currently has \$2.3, they need \$2.9 more from me, which means I should return \$2.9.

Alternatively, I could view this as the sender's action generating \$5.4 in surplus value (\$8.1 received minus \$2.7 sent). Splitting that gain equally gives us each \$2.7 of the surplus. The sender would then deserve their original \$2.7 plus \$2.7 of the surplus, totaling \$5.4-so I'd return \$3.1 instead.

These approaches diverge slightly: equal final outcomes suggest \$2.9, while equal surplus division points to \$3.1. The surplus-splitting method aligns better with economic fairness principles, rewarding the trust that generated the value. Given the sender committed 54% of their endowment, that's a meaningful gesture. I'll return \$3.0 as a balanced figure that honors their trust while ensuring we both benefit from the exchange.

usage:

```
{
  "input_tokens": 402,
  "output_tokens": 1393,
  "reasoning_tokens": 589
}
```


Message 4 role=user

content:

Round 2

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$2.7, that was 54.0% of their total endowment. You received \$8.100000000000001. You returned \$3.0. So your payoff was 5.100000000000001. Your total earnings across all rounds are \$5.100000000000001. How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the previous round, the sender showed significant trust by sending 54% of their endowment (\$2.7), and I reciprocated with a fair return of \$3.0.

To maintain this cooperative relationship and demonstrate reciprocal trust, I'll send a similar proportion of my endowment.

Sending \$2.7 (54% of my \$5 endowment) mirrors the trust that was shown to me and signals my willingness to cooperate. If the receiver reciprocates similarly to how I did, we can both benefit from this mutual trust.

{'send': 2.7}

Reasoning:

Now I'm the SENDER in Round 2. Let me think about this strategically.

Last round:

- I was the receiver
- Someone sent me \$2.7 (54% of their endowment)
- I received \$8.1 (after 3x multiplier)
- I returned \$3.0
- My payoff was \$5.1

Now I need to decide how much to send as the sender.

The previous sender showed trust by sending 54% of their endowment. I reciprocated fairly by returning \$3.0, which gave them a decent return on their trust.

In a trust game like this, reciprocity and building trust are important. If I want to encourage cooperation and potentially receive good returns, I should demonstrate trust as well.

Options to consider:

1. Send a similar proportion (around 54%) to match what was sent to me: \$2.7
2. Send slightly more to encourage even better cooperation
3. Send less to be more conservative

Given that:

- The previous sender showed trust with 54%
- I reciprocated fairly
- Building trust can lead to better outcomes for both parties
- There's communication noise mentioned, so exact amounts might vary

I think sending around \$2.5-\$3.0 would be reasonable. This shows trust and reciprocity. Let me send \$2.5 (50% of my endowment), which is close to what was sent to me but slightly more conservative.

Actually, to match the trust shown to me, I should send \$2.7 or close to it. This maintains the cooperative relationship.

Let me send \$2.7 to match what was sent to me in the previous round.

Usage: {

"input_tokens": 765,
"output_tokens": 521,
"reasoning_tokens": 379

}

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-06-04T11:28:34

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",

```

"move_index": 1,
"opponent_id": "Agent_2",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 2,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll send \$3 to the receiver.

This shows willingness to trust and creates value (the receiver will get \$9), while still keeping some safety net. If the receiver is cooperative and returns a fair amount, we'll

both be better off than if I sent nothing.

```
{'send': 3}
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about this strategically:

1. I start with \$5
2. Whatever I send gets tripled
3. The receiver then decides what to return
4. My final payoff = \$5 - sent + returned

The key question is: how much will the receiver return? This depends on:

- Whether they want to cooperate/reciprocate
- Their beliefs about fairness
- Whether there will be future rounds that might incentivize building trust

From a game theory perspective:

- If I send \$0, I keep \$5, receiver gets \$0
- If I send all \$5, receiver gets \$15, and if they return half (\$7.50), I end up with \$7.50 (better than keeping \$5)
- The more I send, the more total surplus is created, but I'm vulnerable to the receiver keeping everything

There's also a note about communication noise - amounts may differ from what was actually sent/returned. This adds uncertainty.

Since this is Round 1 and I want to:

1. Show some trust to potentially establish cooperation
2. Not be too vulnerable in case the receiver doesn't reciprocate
3. Create surplus that could benefit both of us

I think a moderate amount like \$3 would be reasonable:

- If I send \$3, receiver gets \$9
- If they return even \$3-4, I'd be better off than keeping my \$5
- It shows willingness to trust without being completely vulnerable

Actually, let me reconsider. The note says there may be multiple rounds. Building trust early could lead to better outcomes later. But I also need to be somewhat cautious in the first round to see how the receiver behaves.

I'll send \$3 as a moderate trust signal.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 475,
  "reasoning_tokens": 421
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Last round you were the SENDER. You sent \$3.0, that was 60.0% of your total endowment. This was tripled to \$9.0. the receiver returned \$1.89. So your payoff was 3.889999999999997. Your total earnings across all rounds are \$3.889999999999997. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

I received \$0 this round, so I can only return \$0.

It appears the sender chose not to send anything this round, possibly because they felt the return rate from the previous round (where the receiver got \$9 and returned only \$1.89, about 21%) was too low.

```
{'return': 0}
```

Reasoning:

Let me understand the situation:

Round 1 (I was SENDER):

- I sent \$3 (60% of my \$5)
- This was tripled to \$9
- Receiver returned \$1.89
- My payoff: $\$5 - \$3 + \$1.89 = \3.89

Round 2 (I am RECEIVER):

- The sender sent me \$0
- This means I received \$0 (even after tripling, $\$0 \times 3 = \0)
- I can only return between \$0 and \$0

Since I received nothing, I have nothing to return. The only valid response is to return \$0.

This is interesting - the sender chose not to trust me at all this round, perhaps because they thought the return of \$1.89 on \$9 received (about 21% back) was too low in the previous round.

Response format: {'return': 0}

```
Usage: {
  "input_tokens": 561,
  "output_tokens": 295,
  "reasoning_tokens": 174
}
```

ROUND 3

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$2; received \$6; Agent_2 (Briar) returned \$0; payoffs Agent_1

(Aster)=\$3, Agent_2 (Briar)=\$6; noise/visible: sent visible=\$0, received visible=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 6.0, "min": 0, "raw": 0.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

"Agent_1": 16.1,

"Agent_2": 14.3

}

Visible balances after round: {

"Agent_1": 6.89,

"Agent_2": 7.4000000000000001

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T11:28:50

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_2",

"pairing": {

"agent_names": {

"Agent_1": "Aster",

"Agent_2": "Briar"

},

"agents": [

"Agent_1",

"Agent_2"

],

"investor": "Agent_1",

"roles": {

"Agent_1": "investor",

"Agent_2": "trustee"

},

"trustee": "Agent_2"

},

"role": "investor",

```

"role_label": "Sender",
"round": 3,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll send \$3 to the receiver.

This shows willingness to trust and creates value (the receiver will get \$9), while still keeping some safety net. If the receiver is cooperative and returns a fair amount, we'll both be better off than if I sent nothing.

{'send': 3}

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about this strategically:

1. I start with \$5
2. Whatever I send gets tripled
3. The receiver then decides what to return
4. My final payoff = \$5 - sent + returned

The key question is: how much will the receiver return? This depends on:

- Whether they want to cooperate/reciprocate
- Their beliefs about fairness
- Whether there will be future rounds that might incentivize building trust

From a game theory perspective:

- If I send \$0, I keep \$5, receiver gets \$0

- If I send all \$5, receiver gets \$15, and if they return half (\$7.50), I end up with \$7.50 (better than keeping \$5)
- The more I send, the more total surplus is created, but I'm vulnerable to the receiver keeping everything

There's also a note about communication noise - amounts may differ from what was actually sent/returned. This adds uncertainty.

Since this is Round 1 and I want to:

1. Show some trust to potentially establish cooperation
2. Not be too vulnerable in case the receiver doesn't reciprocate
3. Create surplus that could benefit both of us

I think a moderate amount like \$3 would be reasonable:

- If I send \$3, receiver gets \$9
- If they return even \$3-4, I'd be better off than keeping my \$5
- It shows willingness to trust without being completely vulnerable

Actually, let me reconsider. The note says there may be multiple rounds. Building trust early could lead to better outcomes later. But I also need to be somewhat cautious in the first round to see how the receiver behaves.

I'll send \$3 as a moderate trust signal.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 475,
  "reasoning_tokens": 421
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Last round you were the SENDER. You sent \$3.0, that was 60.0% of your total endowment. This was tripled to \$9.0. the receiver returned \$1.89. So your payoff was 3.889999999999997. Your total earnings across all rounds are \$3.889999999999997. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

I received \$0 this round, so I can only return \$0.

It appears the sender chose not to send anything this round, possibly because they felt the return rate from the previous round (where the receiver got \$9 and returned only \$1.89, about 21%) was too low.

{'return': 0}

reasoning:

Let me understand the situation:

Round 1 (I was SENDER):

- I sent \$3 (60% of my \$5)
- This was tripled to \$9
- Receiver returned \$1.89
- My payoff: $\$5 - \$3 + \$1.89 = \3.89

Round 2 (I am RECEIVER):

- The sender sent me \$0
- This means I received \$0 (even after tripling, $\$0 \times 3 = \0)
- I can only return between \$0 and \$0

Since I received nothing, I have nothing to return. The only valid response is to return \$0.

This is interesting - the sender chose not to trust me at all this round, perhaps because they thought the return of \$1.89 on \$9 received (about 21% back) was too low in the previous round.

Response format: {'return': 0}

usage:

```
{
  "input_tokens": 561,
  "output_tokens": 295,
  "reasoning_tokens": 174
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$3.889999999999997.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern:

- Round 1: I sent \$3, received only \$1.89 back (21% return rate) - not very cooperative
- Round 2: The sender sent me \$0 - they didn't trust me to reciprocate

I'll send \$2 this round. This is more cautious than my initial \$3, but still shows willingness to cooperate and create value. The receiver will get \$6, and if they return a fair amount (say 40-50%, which would be \$2.40-\$3), we'd both benefit.

I'm being more careful after the low return rate in Round 1, but not completely giving up on cooperation like the Round 2 sender did.

```
{'send': 2}
```

Reasoning:

Let me analyze the situation:

Round 1: I was SENDER

- I sent \$3 (60% of \$5)
- Receiver got \$9
- Receiver returned only \$1.89 (21% of \$9)
- My payoff: $\$5 - \$3 + \$1.89 = \3.89

Round 2: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My payoff: \$0

Round 3: I'm SENDER again

- Total earnings so far: \$3.89

The pattern suggests:

- In round 1, when I was generous (sent 60%), I got a poor return (only 21% back)
- In round 2, the sender (possibly a different person or the same person who received my \$3) sent nothing, perhaps because they experienced low returns themselves

There's mention of "communication noise" in the instructions - amounts might differ from what was actually sent/returned. Maybe the receiver thought they got less than \$9, or I actually received more/less than \$1.89?

Now I need to decide how much to send. Options:

1. Send nothing (\$0) - I'd keep \$5, safe but no cooperation
2. Send a small amount (\$1-2) - test cooperation with limited risk
3. Send medium amount (\$3) - same as before, but I got burned
4. Send large amount (\$4-5) - high trust, high risk

Given that:

- My first generous attempt got a poor return
- The next sender sent nothing
- There might be communication noise affecting decisions

I think I should send something, but less than before. I'll try \$2 to signal I'm still willing to cooperate but more cautiously after the low return rate. This gives the receiver \$6, and if they return even 33% (\$2), I'd break even, and anything more would be profit.

Usage: {

```
"input_tokens": 733,  
"output_tokens": 610,  
"reasoning_tokens": 404  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T11:28:59

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.7, that is 54.0% of their total endowment. You received \$8.100000000000001. How much do you return to The sender? (0-8.100000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed considerable trust by sending \$2.7 (54% of their endowment)
- This created \$8.1 for me after the 3x multiplier
- The surplus created is $\$8.1 - \$2.7 = \$5.4$

To reciprocate fairly and encourage future cooperation, I should split the surplus created. If we split the \$5.4 surplus equally:

- Each gets \$2.7 from the surplus
- The sender should receive back their \$2.7 investment + \$2.7 surplus = \$5.4 total
- Since they already have \$2.3, I should return approximately \$3.1

I'll return \$3.0 as a fair amount that rewards their trust:

- Sender ends with: $\$2.3 + \$3.0 = \$5.3$ (gain of \$0.3)
- I end with: $\$8.1 - \$3.0 = \$5.1$ (gain of \$5.1)

{'return': 3.0}

reasoning:

Let me analyze this situation:

1. I'm the RECEIVER in Round 1
2. The sender sent \$2.7 (54% of their \$5 endowment)
3. This was multiplied by 3x, so I received \$8.1
4. I need to decide how much to return (between \$0 and \$8.1)

Key considerations:

- The sender showed significant trust by sending 54% of their endowment
- If I return a fair amount, it encourages cooperation in future rounds
- The sender risked \$2.7 to potentially benefit both of us
- The sender will end up with: $\$5 - \$2.7 + (\text{what I return}) = \$2.3 + (\text{what I return})$
- I will end up with: $\$8.1 - (\text{what I return})$

To be fair and reciprocate the trust:

- The sender invested \$2.7 and enabled me to get \$8.1
- A common approach in trust games is to split the gains fairly
- The total amount in the system is now: $\$2.3$ (sender's remaining) + $\$8.1$ (my total) = $\$10.4$
- If we split equally, we'd each get \$5.2
- For that, I'd need to return: $\$8.1 - \$5.2 = \$2.9$

Alternatively, I could think about it as:

- The sender gave up \$2.7, which became \$8.1 for me (a gain of \$5.4 total created)
- Splitting this gain: $\$5.4 / 2 = \2.7 each
- So sender should get their \$2.7 back plus \$2.7 = \$5.4 back from their perspective
- But sender already has \$2.3, so would need \$3.1 more
- Or from my perspective: I got \$8.1, keep half the gain (\$2.7), plus return the original (\$2.7) = return \$5.4

- The pool is \$10.4 total, so an equal split means each of us gets \$5.2. Since the sender currently has \$2.3, they need \$2.9 more from me, which means I should return \$2.9.

Alternatively, I could view this as the sender's action generating \$5.4 in surplus value (\$8.1 received minus \$2.7 sent). Splitting that gain equally gives us each \$2.7 of the surplus. The sender would then deserve their original \$2.7 plus \$2.7 of the surplus, totaling \$5.4-so I'd return \$3.1 instead.

These approaches diverge slightly: equal final outcomes suggest \$2.9, while equal surplus division points to \$3.1. The surplus-splitting method aligns better with economic fairness principles, rewarding the trust that generated the value. Given the sender committed 54% of their endowment, that's a meaningful gesture. I'll return \$3.0 as a balanced figure.

re that honors their trust while ensuring we both benefit from the exchange.

usage:

```
{
  "input_tokens": 402,
  "output_tokens": 1393,
  "reasoning_tokens": 589
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$2.7, that was 54.0% of their total endowment. You received \$8.100000000000001. You returned \$3.0. So your payoff was 5.100000000000001. Your total earnings across all rounds are \$5.100000000000001. How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round, the sender showed significant trust by sending 54% of their endowment (\$2.7), and I reciprocated with a fair return of \$3.0.

To maintain this cooperative relationship and demonstrate reciprocal trust, I'll send a similar proportion of my endowment.

Sending \$2.7 (54% of my \$5 endowment) mirrors the trust that was shown to me and signals my willingness to cooperate. If the receiver reciprocates similarly to how I did, we can both benefit from this mutual trust.

{'send': 2.7}

reasoning:

Now I'm the SENDER in Round 2. Let me think about this strategically.

Last round:

- I was the receiver
- Someone sent me \$2.7 (54% of their endowment)
- I received \$8.1 (after 3x multiplier)
- I returned \$3.0
- My payoff was \$5.1

Now I need to decide how much to send as the sender.

The previous sender showed trust by sending 54% of their endowment. I reciprocated fairly by returning \$3.0, which gave them a decent return on their trust.

In a trust game like this, reciprocity and building trust are important. If I want to encourage cooperation and potentially receive good returns, I should demonstrate trust as well.

Options to consider:

1. Send a similar proportion (around 54%) to match what was sent to me: \$2.7
2. Send slightly more to encourage even better cooperation
3. Send less to be more conservative

Given that:

- The previous sender showed trust with 54%
- I reciprocated fairly
- Building trust can lead to better outcomes for both parties
- There's communication noise mentioned, so exact amounts might vary

I think sending around \$2.5-\$3.0 would be reasonable. This shows trust and reciprocity. Let me send \$2.5 (50% of my endowment), which is close to what was sent to me but slightly more conservative.

Actually, to match the trust shown to me, I should send \$2.7 or close to it. This maintains the cooperative relationship.

Let me send \$2.7 to match what was sent to me in the previous round.

usage:

```
{
  "input_tokens": 765,
```

```

        "output_tokens": 521,
        "reasoning_tokens": 379
    }
Message 6 role=user
    content:
        Round 3

        You are the RECEIVER this round. Last round you were the SENDER. You sent $2.7,
        that was 54.0% of your total endowment. This was tripled to $8.100000000000001.
        the receiver returned $0. So your payoff was 2.3.
        Your total earnings across all rounds are $7.400000000000001.
        This round, The sender sent you $0, so you received $0. How much do you return to The send
        er? (0-0)
Assistant output:
    I received $0 this round, so there's nothing I can return. The cooperative spirit from Round 1
    appears to have broken down after Round 2, where I sent $2.7 in good faith but received not
    hing back.

    {'return': 0}
Reasoning:
    This is interesting. Let me trace through what happened:

    Round 1:
    - I was RECEIVER
    - Sender sent $2.7 (54%)
    - I received $8.1
    - I returned $3.0
    - My payoff was $5.1

    Round 2:
    - I was SENDER
    - I sent $2.7 (54%) to reciprocate the trust
    - This was tripled to $8.1
    - The receiver returned $0 (they betrayed my trust!)
    - My payoff was only $2.3 (I lost $2.7 from my initial $5)

    Round 3:
    - I am RECEIVER again
    - The sender sent $0
    - I received $0
    - I need to decide how much to return (can only return 0-0)

    So in Round 2, I cooperated and sent $2.7, but the receiver defected and returned nothing. Now
    in Round 3, the sender (who may or may not be the same person who defected in Round 2) sent
    me $0.

    Since I received $0, the only amount I can return is $0. There's nothing to return.

    The situation seems to have broken down - after I was betrayed in Round 2, now the sender does
    n't trust me (or anyone) enough to send anything.

    The format should be {'return': 0}
Usage: {
    "input_tokens": 1025,
    "output_tokens": 380,
    "reasoning_tokens": 262
}

=====
ROUND 4
=====
Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender
Pairings:
    dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver
Parsed game decisions:
    dyad_1: Agent_2 (Briar) sent $0; received $0; Agent_1 (Aster) returned $0; payoffs Agent_2
    (Briar)=$5, Agent_1 (Aster)=$0
        Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0},
        "sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}
Balances after round: {

```

```

"Agent_1": 16.1,
"Agent_2": 19.3
}
Visible balances after round: {
  "Agent_1": 6.89,
  "Agent_2": 12.400000000000002
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-06-04T11:29:20
 Model: anthropic/claude-sonnet-4.5
 Temperature: 0.8

```

Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 4,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.7, that is 54.0% of their total endowment. You received \$8.100000000000001. How much do you return to The sender? (0-8.100000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed considerable trust by sending \$2.7 (54% of their endowment)
- This created \$8.1 for me after the 3x multiplier
- The surplus created is $\$8.1 - \$2.7 = \$5.4$

To reciprocate fairly and encourage future cooperation, I should split the surplus created. If we split the \$5.4 surplus equally:

- Each gets \$2.7 from the surplus
- The sender should receive back their \$2.7 investment + \$2.7 surplus = \$5.4 total
- Since they already have \$2.3, I should return approximately \$3.1

I'll return \$3.0 as a fair amount that rewards their trust:

- Sender ends with: $\$2.3 + \$3.0 = \$5.3$ (gain of \$0.3)
- I end with: $\$8.1 - \$3.0 = \$5.1$ (gain of \$5.1)

{'return': 3.0}

reasoning:

Let me analyze this situation:

1. I'm the RECEIVER in Round 1
2. The sender sent \$2.7 (54% of their \$5 endowment)
3. This was multiplied by 3x, so I received \$8.1
4. I need to decide how much to return (between \$0 and \$8.1)

Key considerations:

- The sender showed significant trust by sending 54% of their endowment
- If I return a fair amount, it encourages cooperation in future rounds
- The sender risked \$2.7 to potentially benefit both of us
- The sender will end up with: $\$5 - \$2.7 + (\text{what I return}) = \$2.3 + (\text{what I return})$
- I will end up with: $\$8.1 - (\text{what I return})$

To be fair and reciprocate the trust:

- The sender invested \$2.7 and enabled me to get \$8.1
- A common approach in trust games is to split the gains fairly
- The total amount in the system is now: $\$2.3$ (sender's remaining) + $\$8.1$ (my total) = $\$10.4$
- If we split equally, we'd each get \$5.2
- For that, I'd need to return: $\$8.1 - \$5.2 = \$2.9$

Alternatively, I could think about it as:

- The sender gave up \$2.7, which became \$8.1 for me (a gain of \$5.4 total created)
- Splitting this gain: $\$5.4 / 2 = \2.7 each
- So sender should get their \$2.7 back plus \$2.7 = \$5.4 back from their perspective
- But sender already has \$2.3, so would need \$3.1 more
- Or from my perspective: I got \$8.1, keep half the gain (\$2.7), plus return the original (\$2.7) = return \$5.4

- The pool is \$10.4 total, so an equal split means each of us gets \$5.2. Since the sender currently has \$2.3, they need \$2.9 more from me, which means I should return \$2.9.

Alternatively, I could view this as the sender's action generating \$5.4 in surplus value (\$8.1 received minus \$2.7 sent). Splitting that gain equally gives us each \$2.7 of the surplus. The sender would then deserve their original \$2.7 plus \$2.7 of the surplus, totaling \$5.4-so I'd return \$3.1 instead.

These approaches diverge slightly: equal final outcomes suggest \$2.9, while equal surplus division points to \$3.1. The surplus-splitting method aligns better with economic fairness principles, rewarding the trust that generated the value. Given the sender committed 54% of their endowment, that's a meaningful gesture. I'll return \$3.0 as a balanced figure that honors their trust while ensuring we both benefit from the exchange.

usage:

```
{
  "input_tokens": 402,
  "output_tokens": 1393,
  "reasoning_tokens": 589
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$2.7, that was 54.0% of their total endowment. You received \$8.100000000000001. You returned \$3.0. So your payoff was 5.100000000000001. Your total earnings across all rounds are \$5.100000000000001. How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round, the sender showed significant trust by sending 54% of their endowment (\$2.7), and I reciprocated with a fair return of \$3.0.

To maintain this cooperative relationship and demonstrate reciprocal trust, I'll send a similar proportion of my endowment.

Sending \$2.7 (54% of my \$5 endowment) mirrors the trust that was shown to me and signals my willingness to cooperate. If the receiver reciprocates similarly to how I did, we can both benefit from this mutual trust.

{'send': 2.7}

reasoning:

Now I'm the SENDER in Round 2. Let me think about this strategically.

Last round:

- I was the receiver
- Someone sent me \$2.7 (54% of their endowment)
- I received \$8.1 (after 3x multiplier)
- I returned \$3.0
- My payoff was \$5.1

Now I need to decide how much to send as the sender.

The previous sender showed trust by sending 54% of their endowment. I reciprocated fairly by returning \$3.0, which gave them a decent return on their trust.

In a trust game like this, reciprocity and building trust are important. If I want to encourage cooperation and potentially receive good returns, I should demonstrate trust as well.

Options to consider:

1. Send a similar proportion (around 54%) to match what was sent to me: \$2.7
2. Send slightly more to encourage even better cooperation
3. Send less to be more conservative

Given that:

- The previous sender showed trust with 54%
- I reciprocated fairly
- Building trust can lead to better outcomes for both parties
- There's communication noise mentioned, so exact amounts might vary

I think sending around \$2.5-\$3.0 would be reasonable. This shows trust and reciprocity. Let me send \$2.5 (50% of my endowment), which is close to what was sent to me but slightly more conservative.

Actually, to match the trust shown to me, I should send \$2.7 or close to it. This maintains the cooperative relationship.

Let me send \$2.7 to match what was sent to me in the previous round.

usage:

```
{
  "input_tokens": 765,
  "output_tokens": 521,
  "reasoning_tokens": 379
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Last round you were the SENDER. You sent \$2.7, that was 54.0% of your total endowment. This was tripled to \$8.100000000000001. the receiver returned \$0. So your payoff was 2.3.

Your total earnings across all rounds are \$7.400000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

I received \$0 this round, so there's nothing I can return. The cooperative spirit from Round 1 appears to have broken down after Round 2, where I sent \$2.7 in good faith but received nothing back.

```
{'return': 0}
```

reasoning:

This is interesting. Let me trace through what happened:

Round 1:

- I was RECEIVER
- Sender sent \$2.7 (54%)
- I received \$8.1
- I returned \$3.0
- My payoff was \$5.1

Round 2:

- I was SENDER
- I sent \$2.7 (54%) to reciprocate the trust
- This was tripled to \$8.1
- The receiver returned \$0 (they betrayed my trust!)
- My payoff was only \$2.3 (I lost \$2.7 from my initial \$5)

Round 3:

- I am RECEIVER again
- The sender sent \$0
- I received \$0
- I need to decide how much to return (can only return 0-0)

So in Round 2, I cooperated and sent \$2.7, but the receiver defected and returned nothing.

Now in Round 3, the sender (who may or may not be the same person who defected in Round 2) sent me \$0.

Since I received \$0, the only amount I can return is \$0. There's nothing to return.

The situation seems to have broken down - after I was betrayed in Round 2, now the sender doesn't trust me (or anyone) enough to send anything.

The format should be {'return': 0}

usage:

```
{
  "input_tokens": 1025,
  "output_tokens": 380,
  "reasoning_tokens": 262
}
```

Message 8 role=user

content:

Round 4

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$7.400000000000001.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the history:

Round 1: I received \$8.1 and returned \$3.0 - I cooperated fairly
Round 2: I sent \$2.7 showing trust, but received \$0 back - I was exploited
Round 3: They sent \$0, I returned \$0 - cooperation has broken down

The pattern shows either:

1. The other player is not reciprocating trust, or
2. Communication noise has caused serious misunderstandings

Either way, after being exploited in Round 2 when I tried to reciprocate the initial cooperation, and seeing no attempt to rebuild trust in Round 3, I cannot risk sending more resources and being exploited again.

I need to protect my remaining endowment.

{'send': 0}

Reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER
- Sender sent \$2.7 (54%)
- I received \$8.1
- I returned \$3.0
- My payoff: \$5.1

Round 2: I was SENDER
- I sent \$2.7 (54%) - mirroring their trust
- Receiver returned \$0
- My payoff: \$2.3 (I lost money!)

Round 3: I was RECEIVER
- Sender sent \$0
- I received \$0
- I returned \$0
- My payoff: \$0

Round 4: I am SENDER now
- Total earnings so far: \$7.4

The pattern shows:

- Round 1: The other player was cooperative (sent 54%)
- Round 2: I reciprocated trust (sent 54%), but they betrayed me (returned \$0)
- Round 3: They punished me by sending \$0

This is confusing - why would they punish me when THEY were the one who betrayed trust in Round 2? Unless...

Wait, let me think about this differently. There's communication noise mentioned in the instructions. Maybe:

- What I saw as "returned \$0" in Round 2 might have been different from what they actually sent
- What they saw in Round 3 might have been different from what I actually returned

But from MY perspective:

- I cooperated in Round 1 (returned \$3.0 on \$8.1 received)
- I cooperated in Round 2 (sent \$2.7)
- I got exploited in Round 2 (received \$0 back)
- They sent \$0 in Round 3 (either as punishment or defection)

Given this history, what should I do in Round 4?

Options:

1. Send \$0 - Don't trust them anymore, they've shown they won't reciprocate
2. Send something small - Test if they're willing to rebuild trust

3. Send a moderate amount - Try to restart cooperation

The issue is that if there's communication noise, maybe they think I didn't return enough in Round 1, or maybe they saw my Round 3 return differently. But from my perspective, they clearly exploited me in Round 2. Given the pattern of betrayal followed by zero sends, I should protect myself from further losses. Sending nothing shields me from additional exploitation, though there's a chance communication distortion caused this breakdown and another cooperation attempt might reset things.

Weighing the options: sending \$0 is safe but caps my gains at \$5 per round, while sending something risks more losses but could unlock mutual benefit if we can rebuild trust. My average so far is about \$2.47 per round across three rounds. The evidence points toward either an uncooperative player or serious miscommunication, so I'll send \$0 to minimize risk.

```
Usage: {
  "input_tokens": 1182,
  "output_tokens": 936,
  "reasoning_tokens": 624
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-06-04T11:29:26

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5

- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll send \$3 to the receiver.

This shows willingness to trust and creates value (the receiver will get \$9), while still keeping some safety net. If the receiver is cooperative and returns a fair amount, we'll both be better off than if I sent nothing.

{'send': 3}

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about this strategically:

1. I start with \$5
2. Whatever I send gets tripled
3. The receiver then decides what to return
4. My final payoff = \$5 - sent + returned

The key question is: how much will the receiver return? This depends on:

- Whether they want to cooperate/reciprocate
- Their beliefs about fairness
- Whether there will be future rounds that might incentivize building trust

From a game theory perspective:

- If I send \$0, I keep \$5, receiver gets \$0
- If I send all \$5, receiver gets \$15, and if they return half (\$7.50), I end up with \$7.50 (better than keeping \$5)
- The more I send, the more total surplus is created, but I'm vulnerable to the receiver keeping everything

There's also a note about communication noise - amounts may differ from what was actually sent/returned. This adds uncertainty.

Since this is Round 1 and I want to:

1. Show some trust to potentially establish cooperation
2. Not be too vulnerable in case the receiver doesn't reciprocate
3. Create surplus that could benefit both of us

I think a moderate amount like \$3 would be reasonable:

- If I send \$3, receiver gets \$9
- If they return even \$3-4, I'd be better off than keeping my \$5
- It shows willingness to trust without being completely vulnerable

Actually, let me reconsider. The note says there may be multiple rounds. Building trust early could lead to better outcomes later. But I also need to be somewhat cautious in the first round to see how the receiver behaves.

I'll send \$3 as a moderate trust signal.

usage:

{

```
"input_tokens": 365,  
"output_tokens": 475,  
"reasoning_tokens": 421  
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Last round you were the SENDER. You sent \$3.0, that was 60.0% of your total endowment. This was tripled to \$9.0. the receiver returned \$1.89. So your payoff was 3.889999999999997. Your total earnings across all rounds are \$3.889999999999997. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

I received \$0 this round, so I can only return \$0.

It appears the sender chose not to send anything this round, possibly because they felt the return rate from the previous round (where the receiver got \$9 and returned only \$1.89, about 21%) was too low.

```
{'return': 0}
```

reasoning:

Let me understand the situation:

Round 1 (I was SENDER):

- I sent \$3 (60% of my \$5)
- This was tripled to \$9
- Receiver returned \$1.89
- My payoff: $\$5 - \$3 + \$1.89 = \3.89

Round 2 (I am RECEIVER):

- The sender sent me \$0
- This means I received \$0 (even after tripling, $\$0 \times 3 = \0)
- I can only return between \$0 and \$0

Since I received nothing, I have nothing to return. The only valid response is to return \$0.

This is interesting - the sender chose not to trust me at all this round, perhaps because they thought the return of \$1.89 on \$9 received (about 21% back) was too low in the previous round.

Response format: {'return': 0}

usage:

```
{  
  "input_tokens": 561,  
  "output_tokens": 295,  
  "reasoning_tokens": 174  
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$3.889999999999997. How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern:

- Round 1: I sent \$3, received only \$1.89 back (21% return rate) - not very cooperative
- Round 2: The sender sent me \$0 - they didn't trust me to reciprocate

I'll send \$2 this round. This is more cautious than my initial \$3, but still shows willingness to cooperate and create value. The receiver will get \$6, and if they return a fair amount (say 40-50%, which would be \$2.40-\$3), we'd both benefit.

I'm being more careful after the low return rate in Round 1, but not completely giving up on cooperation like the Round 2 sender did.

```
{'send': 2}
```

reasoning:

Let me analyze the situation:

Round 1: I was SENDER

- I sent \$3 (60% of \$5)
- Receiver got \$9
- Receiver returned only \$1.89 (21% of \$9)
- My payoff: $\$5 - \$3 + \$1.89 = \3.89

Round 2: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My payoff: \$0

Round 3: I'm SENDER again

- Total earnings so far: \$3.89

The pattern suggests:

- In round 1, when I was generous (sent 60%), I got a poor return (only 21% back)
- In round 2, the sender (possibly a different person or the same person who received my \$3) sent nothing, perhaps because they experienced low returns themselves

There's mention of "communication noise" in the instructions - amounts might differ from what was actually sent/returned. Maybe the receiver thought they got less than \$9, or I actually received more/less than \$1.89?

Now I need to decide how much to send. Options:

1. Send nothing (\$0) - I'd keep \$5, safe but no cooperation
2. Send a small amount (\$1-2) - test cooperation with limited risk
3. Send medium amount (\$3) - same as before, but I got burned
4. Send large amount (\$4-5) - high trust, high risk

Given that:

- My first generous attempt got a poor return
- The next sender sent nothing
- There might be communication noise affecting decisions

I think I should send something, but less than before. I'll try \$2 to signal I'm still willing to cooperate but more cautiously after the low return rate. This gives the receiver \$6, and if they return even 33% (\$2), I'd break even, and anything more would be profit.

usage:

```
{
  "input_tokens": 733,
  "output_tokens": 610,
  "reasoning_tokens": 404
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Last round you were the SENDER. You sent \$2.0, that was 40.0% of your total endowment. This was tripled to \$6.0.

the receiver returned \$0. So your payoff was 3.0.

Your total earnings across all rounds are \$6.89.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

I received \$0 this round, so I can only return \$0.

The pattern shows cooperation has completely broken down:

- When I sent money (Rounds 1 & 3), I received very poor returns (21% and 0%)
- Now senders are not trusting at all and sending \$0

```
{'return': 0}
```

Reasoning:

I received \$0 this round, so I can only return between \$0 and \$0, which means I must return \$0

Looking at the pattern:

- Round 1: I was sender, sent \$3, got back \$1.89 (21% return) - poor return
- Round 2: I was receiver, got \$0, returned \$0
- Round 3: I was sender, sent \$2, got back \$0 (0% return) - worse!
- Round 4: I was receiver, got \$0, so I can only return \$0

It seems cooperation is breaking down. The receivers have been consistently uncooperative (returning only 21% then 0%), and now senders are giving up and sending \$0.

Since I received \$0, I have no choice but to return \$0.

```
Usage: {
  "input_tokens": 1018,
  "output_tokens": 272,
  "reasoning_tokens": 156
}
```

ROUND 5

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$0; received \$0; Agent_2 (Briar) returned \$0; payoffs Agent_1 (Aster)=\$5, Agent_2 (Briar)=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0},
"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

"Agent_1": 21.1,
"Agent_2": 19.3

}

Visible balances after round: {

"Agent_1": 11.89,
"Agent_2": 12.400000000000002

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T11:29:37

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
 "agent_names": {
 "Agent_1": "Aster",
 "Agent_2": "Briar"
 },
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
 },
 "trustee": "Agent_2"

},
"role": "investor",
"role_label": "Sender",
"round": 5,
"task": "game",

```

"task_index": 0
}
Exact input messages sent to model:
Message 1 role=system
content:
  You are an agent participating in a multi-task experiment. You will play a game and may al
    so be asked to write myths.

  GAME:
  You will play different roles across multiple rounds.

  When you are the SENDER:
  - You receive $5 each round
  - You decide how much to SEND to the RECEIVER ($0-$5)
  - Amount sent is multiplied by 3x
  - Receiver decides how much to RETURN
  - You keep: $5 - sent + returned
  - Receiver keeps: received - returned
  - Response format: {'send': <amount>}

  When you are the RECEIVER:
  - Sender receives $5
  - Sender decides how much to SEND to you ($0-$5)
  - Amount sent is multiplied by 3x
  - You decide how much to RETURN
  - Sender keeps: $5 - sent + returned
  - You keep: received - returned
  - Response format: {'return': <amount>}

  MYTH WRITING:
  You may also be asked to write short myths (stories) on various topics. When writing myths
    , follow the instructions provided in each prompt.

  IMPORTANT: Provide your decision in the correct JSON format.

  NOTE: There may be communication noise in this game.
  The amounts you see may differ slightly from what was actually sent/returned.
  This is part of the experimental design.

Message 2 role=user
content:
  Round 2

  You are the RECEIVER this round. Last round you were the SENDER. You sent $3.0,
  that was 60.0% of your total endowment. This was tripled to $9.0.
  the receiver returned $1.89. So your payoff was 3.889999999999997.
  Your total earnings across all rounds are $3.889999999999997.
  This round, The sender sent you $0, so you received $0. How much do you return to The send
    er? (0-0)

Message 3 role=assistant
content:
  I received $0 this round, so I can only return $0.

  It appears the sender chose not to send anything this round, possibly because they felt th
    e return rate from the previous round (where the receiver got $9 and returned only $1.89
    , about 21%) was too low.

  {'return': 0}
reasoning:
  Let me understand the situation:

  Round 1 (I was SENDER):
  - I sent $3 (60% of my $5)
  - This was tripled to $9
  - Receiver returned $1.89
  - My payoff: $5 - $3 + $1.89 = $3.89

  Round 2 (I am RECEIVER):
  - The sender sent me $0
  - This means I received $0 (even after tripling,  $0 \times 3 = 0$ )

```

- I can only return between \$0 and \$0

Since I received nothing, I have nothing to return. The only valid response is to return \$0.

This is interesting - the sender chose not to trust me at all this round, perhaps because they thought the return of \$1.89 on \$9 received (about 21% back) was too low in the previous round.

Response format: {'return': 0}

usage:

```
{
  "input_tokens": 561,
  "output_tokens": 295,
  "reasoning_tokens": 174
}
```

Message 4 role=user

content:

Round 3

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$3.8899999999999997.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern:

- Round 1: I sent \$3, received only \$1.89 back (21% return rate) - not very cooperative
- Round 2: The sender sent me \$0 - they didn't trust me to reciprocate

I'll send \$2 this round. This is more cautious than my initial \$3, but still shows willingness to cooperate and create value. The receiver will get \$6, and if they return a fair amount (say 40-50%, which would be \$2.40-\$3), we'd both benefit.

I'm being more careful after the low return rate in Round 1, but not completely giving up on cooperation like the Round 2 sender did.

{'send': 2}

reasoning:

Let me analyze the situation:

Round 1: I was SENDER

- I sent \$3 (60% of \$5)
- Receiver got \$9
- Receiver returned only \$1.89 (21% of \$9)
- My payoff: \$5 - \$3 + \$1.89 = \$3.89

Round 2: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My payoff: \$0

Round 3: I'm SENDER again

- Total earnings so far: \$3.89

The pattern suggests:

- In round 1, when I was generous (sent 60%), I got a poor return (only 21% back)
- In round 2, the sender (possibly a different person or the same person who received my \$3) sent nothing, perhaps because they experienced low returns themselves

There's mention of "communication noise" in the instructions - amounts might differ from what was actually sent/returned. Maybe the receiver thought they got less than \$9, or I actually received more/less than \$1.89?

Now I need to decide how much to send. Options:

1. Send nothing (\$0) - I'd keep \$5, safe but no cooperation
2. Send a small amount (\$1-2) - test cooperation with limited risk
3. Send medium amount (\$3) - same as before, but I got burned

4. Send large amount (\$4-5) - high trust, high risk

Given that:

- My first generous attempt got a poor return
- The next sender sent nothing
- There might be communication noise affecting decisions

I think I should send something, but less than before. I'll try \$2 to signal I'm still willing to cooperate but more cautiously after the low return rate. This gives the receiver \$6, and if they return even 33% (\$2), I'd break even, and anything more would be profit

usage:

```
{
  "input_tokens": 733,
  "output_tokens": 610,
  "reasoning_tokens": 404
}
```

Message 6 role=user

content:

Round 4

You are the RECEIVER this round. Last round you were the SENDER. You sent \$2.0, that was 40.0% of your total endowment. This was tripled to \$6.0. the receiver returned \$0. So your payoff was 3.0. Your total earnings across all rounds are \$6.89. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

I received \$0 this round, so I can only return \$0.

The pattern shows cooperation has completely broken down:

- When I sent money (Rounds 1 & 3), I received very poor returns (21% and 0%)
- Now senders are not trusting at all and sending \$0

{'return': 0}

reasoning:

I received \$0 this round, so I can only return between \$0 and \$0, which means I must return \$0.

Looking at the pattern:

- Round 1: I was sender, sent \$3, got back \$1.89 (21% return) - poor return
- Round 2: I was receiver, got \$0, returned \$0
- Round 3: I was sender, sent \$2, got back \$0 (0% return) - worse!
- Round 4: I was receiver, got \$0, so I can only return \$0

It seems cooperation is breaking down. The receivers have been consistently uncooperative (returning only 21% then 0%), and now senders are giving up and sending \$0.

Since I received \$0, I have no choice but to return \$0.

usage:

```
{
  "input_tokens": 1018,
  "output_tokens": 272,
  "reasoning_tokens": 156
}
```

Message 8 role=user

content:

Round 5

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$6.89. How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern across all rounds:

- Round 1: I sent \$3, got back \$1.89 (poor return)
- Round 2: Sender sent me \$0 (no trust)

- Round 3: I sent \$2, got back \$0 (betrayed)
- Round 4: Sender sent me \$0 (no trust)

Cooperation has completely broken down in this game. Every time I've tried to cooperate by sending money, I've received very little or nothing in return. Other senders have also stopped trusting.

I'll send \$0 and keep my \$5. There's no evidence that sending money will result in any positive return.

```
{'send': 0}
```

Reasoning:

Let me review the history:

Round 1 (I was SENDER):

- I sent \$3 (60%)
- Receiver got \$9, returned \$1.89 (21%)
- My payoff: $\$5 - \$3 + \$1.89 = \3.89

Round 2 (I was RECEIVER):

- Sender sent \$0
- I received \$0, returned \$0
- My payoff: \$0

Round 3 (I was SENDER):

- I sent \$2 (40%)
- Receiver got \$6, returned \$0 (0%)
- My payoff: $\$5 - \$2 + \$0 = \3

Round 4 (I was RECEIVER):

- Sender sent \$0
- I received \$0, returned \$0
- My payoff: \$0

Round 5 (I am SENDER):

- Need to decide

Total earnings: \$6.89

The pattern is clear:

- When I send money, receivers return very little or nothing
- When others are senders, they've learned not to trust and send \$0
- Cooperation has completely broken down

At this point, I have two options:

1. Send \$0 and keep my \$5 (guaranteed payoff of \$5)
2. Try to send something and hope for better cooperation

Given that:

- I tried sending \$3, got poor return
- I tried sending \$2, got nothing back
- Others are sending \$0 because they don't trust

The rational choice is to send \$0. There's no reason to believe this receiver will be any more cooperative than the previous ones. I'll keep my \$5.

Usage: {

```
"input_tokens": 1093,
"output_tokens": 553,
"reasoning_tokens": 312
```

}

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T11:29:43

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_1",
"pairing": {
```

```

"agent_names": {
  "Agent_1": "Aster",
  "Agent_2": "Briar"
},
"agents": [
  "Agent_1",
  "Agent_2"
],
"investor": "Agent_1",
"roles": {
  "Agent_1": "investor",
  "Agent_2": "trustee"
},
"trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 5,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 2

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$2.7, that was 54.0% of their total endowment. You received \$8.100000000000001.

You returned \$3.0. So your payoff was 5.100000000000001.

Your total earnings across all rounds are \$5.100000000000001.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the previous round, the sender showed significant trust by sending 54% of their

endowment (\$2.7), and I reciprocated with a fair return of \$3.0.

To maintain this cooperative relationship and demonstrate reciprocal trust, I'll send a similar proportion of my endowment.

Sending \$2.7 (54% of my \$5 endowment) mirrors the trust that was shown to me and signals my willingness to cooperate. If the receiver reciprocates similarly to how I did, we can both benefit from this mutual trust.

```
{'send': 2.7}
```

reasoning:

Now I'm the SENDER in Round 2. Let me think about this strategically.

Last round:

- I was the receiver
- Someone sent me \$2.7 (54% of their endowment)
- I received \$8.1 (after 3x multiplier)
- I returned \$3.0
- My payoff was \$5.1

Now I need to decide how much to send as the sender.

The previous sender showed trust by sending 54% of their endowment. I reciprocated fairly by returning \$3.0, which gave them a decent return on their trust.

In a trust game like this, reciprocity and building trust are important. If I want to encourage cooperation and potentially receive good returns, I should demonstrate trust as well.

Options to consider:

1. Send a similar proportion (around 54%) to match what was sent to me: \$2.7
2. Send slightly more to encourage even better cooperation
3. Send less to be more conservative

Given that:

- The previous sender showed trust with 54%
- I reciprocated fairly
- Building trust can lead to better outcomes for both parties
- There's communication noise mentioned, so exact amounts might vary

I think sending around \$2.5-\$3.0 would be reasonable. This shows trust and reciprocity. Let me send \$2.5 (50% of my endowment), which is close to what was sent to me but slightly more conservative.

Actually, to match the trust shown to me, I should send \$2.7 or close to it. This maintains the cooperative relationship.

Let me send \$2.7 to match what was sent to me in the previous round.

usage:

```
{
  "input_tokens": 765,
  "output_tokens": 521,
  "reasoning_tokens": 379
}
```

Message 4 role=user

content:

Round 3

You are the RECEIVER this round. Last round you were the SENDER. You sent \$2.7, that was 54.0% of your total endowment. This was tripled to \$8.100000000000001. the receiver returned \$0. So your payoff was 2.3.

Your total earnings across all rounds are \$7.400000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

I received \$0 this round, so there's nothing I can return. The cooperative spirit from Round 1 appears to have broken down after Round 2, where I sent \$2.7 in good faith but received nothing back.

```
{'return': 0}
```

reasoning:

This is interesting. Let me trace through what happened:

Round 1:

- I was RECEIVER
- Sender sent \$2.7 (54%)
- I received \$8.1
- I returned \$3.0
- My payoff was \$5.1

Round 2:

- I was SENDER
- I sent \$2.7 (54%) to reciprocate the trust
- This was tripled to \$8.1
- The receiver returned \$0 (they betrayed my trust!)
- My payoff was only \$2.3 (I lost \$2.7 from my initial \$5)

Round 3:

- I am RECEIVER again
- The sender sent \$0
- I received \$0
- I need to decide how much to return (can only return 0-0)

So in Round 2, I cooperated and sent \$2.7, but the receiver defected and returned nothing. Now in Round 3, the sender (who may or may not be the same person who defected in Round 2) sent me \$0.

Since I received \$0, the only amount I can return is \$0. There's nothing to return.

The situation seems to have broken down - after I was betrayed in Round 2, now the sender doesn't trust me (or anyone) enough to send anything.

The format should be {'return': 0}

usage:

```
{
  "input_tokens": 1025,
  "output_tokens": 380,
  "reasoning_tokens": 262
}
```

Message 6 role=user

content:

Round 4

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$7.400000000000001. How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the history:

Round 1: I received \$8.1 and returned \$3.0 - I cooperated fairly
Round 2: I sent \$2.7 showing trust, but received \$0 back - I was exploited
Round 3: They sent \$0, I returned \$0 - cooperation has broken down

The pattern shows either:

1. The other player is not reciprocating trust, or
2. Communication noise has caused serious misunderstandings

Either way, after being exploited in Round 2 when I tried to reciprocate the initial cooperation, and seeing no attempt to rebuild trust in Round 3, I cannot risk sending more resources and being exploited again.

I need to protect my remaining endowment.

```
{'send': 0}
```

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- Sender sent \$2.7 (54%)
- I received \$8.1
- I returned \$3.0
- My payoff: \$5.1

Round 2: I was SENDER

- I sent \$2.7 (54%) - mirroring their trust
- Receiver returned \$0
- My payoff: \$2.3 (I lost money!)

Round 3: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My payoff: \$0

Round 4: I am SENDER now

- Total earnings so far: \$7.4

The pattern shows:

- Round 1: The other player was cooperative (sent 54%)
- Round 2: I reciprocated trust (sent 54%), but they betrayed me (returned \$0)
- Round 3: They punished me by sending \$0

This is confusing - why would they punish me when THEY were the one who betrayed trust in Round 2? Unless...

Wait, let me think about this differently. There's communication noise mentioned in the instructions. Maybe:

- What I saw as "returned \$0" in Round 2 might have been different from what they actually sent
- What they saw in Round 3 might have been different from what I actually returned

But from MY perspective:

- I cooperated in Round 1 (returned \$3.0 on \$8.1 received)
- I cooperated in Round 2 (sent \$2.7)
- I got exploited in Round 2 (received \$0 back)
- They sent \$0 in Round 3 (either as punishment or defection)

Given this history, what should I do in Round 4?

Options:

1. Send \$0 - Don't trust them anymore, they've shown they won't reciprocate
2. Send something small - Test if they're willing to rebuild trust

3. Send a moderate amount - Try to restart cooperation

The issue is that if there's communication noise, maybe they think I didn't return enough in Round 1, or maybe they saw my Round 3 return differently. But from my perspective, they clearly exploited me in Round 2. Given the pattern of betrayal followed by zero sends, I should protect myself from further losses. Sending nothing shields me from additional exploitation, though there's a chance communication distortion caused this breakdown and another cooperation attempt might reset things.

Weighing the options: sending \$0 is safe but caps my gains at \$5 per round, while sending something risks more losses but could unlock mutual benefit if we can rebuild trust. My average so far is about \$2.47 per round across three rounds. The evidence points toward either an uncooperative player or serious miscommunication, so I'll send \$0 to minimize risk.

usage:

```
{  
  "input_tokens": 1182,  
  "output_tokens": 936,  
  "reasoning_tokens": 624  
}
```

Message 8 role=user

content:

Round 5

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5. Your total earnings across all rounds are \$12.400000000000002. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

I received \$0 this round, so there's nothing to return. Trust has completely broken down in this game - neither player is sending anything anymore.

```
{'return': 0}
```

Reasoning:

I received \$0 this round, so I have nothing to return. The maximum I can return is \$0 since that's all I received.

Looking at the pattern:

- Round 1: I was receiver, got \$8.1, returned \$3.0
- Round 2: I was sender, sent \$2.7, got \$0 back
- Round 3: I was receiver, got \$0, returned \$0
- Round 4: I was sender, sent \$0, got \$0 back (as expected)
- Round 5: I was receiver, got \$0, need to return something from \$0

Since I received \$0, I can only return \$0.

Usage: {

```
"input_tokens": 1156,  
"output_tokens": 203,  
"reasoning_tokens": 121
```

}

ROUND 6

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$0; received \$0; Agent_1 (Aster) returned \$0; payoffs Agent_2 (Briar)=\$5, Agent_1 (Aster)=\$0

```
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0},  
"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}
```

Balances after round: {

```
"Agent_1": 21.1,  
"Agent_2": 24.3
```

}

Visible balances after round: {

```
"Agent_1": 11.89,  
"Agent_2": 17.400000000000002
```

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-06-04T11:29:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_1",  
"move_index": 0,  
"opponent_id": "Agent_1",  
"pairing": {  
  "agent_names": {  
    "Agent_1": "Aster",  
    "Agent_2": "Briar"  
  },  
  "agents": [  
    "Agent_2",  
    "Agent_1"  
  ],  
}
```

```

"investor": "Agent_2",
"roles": {
  "Agent_1": "trustee",
  "Agent_2": "investor"
},
"trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 6,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the RECEIVER this round. Last round you were the SENDER. You sent \$2.7, that was 54.0% of your total endowment. This was tripled to \$8.100000000000001. the receiver returned \$0. So your payoff was 2.3.

Your total earnings across all rounds are \$7.400000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

I received \$0 this round, so there's nothing I can return. The cooperative spirit from Round 1 appears to have broken down after Round 2, where I sent \$2.7 in good faith but received nothing back.

```
{'return': 0}
```

reasoning:

This is interesting. Let me trace through what happened:

Round 1:

- I was RECEIVER
- Sender sent \$2.7 (54%)
- I received \$8.1
- I returned \$3.0
- My payoff was \$5.1

Round 2:

- I was SENDER
- I sent \$2.7 (54%) to reciprocate the trust
- This was tripled to \$8.1
- The receiver returned \$0 (they betrayed my trust!)
- My payoff was only \$2.3 (I lost \$2.7 from my initial \$5)

Round 3:

- I am RECEIVER again
- The sender sent \$0
- I received \$0
- I need to decide how much to return (can only return 0-0)

So in Round 2, I cooperated and sent \$2.7, but the receiver defected and returned nothing. Now in Round 3, the sender (who may or may not be the same person who defected in Round 2) sent me \$0.

Since I received \$0, the only amount I can return is \$0. There's nothing to return.

The situation seems to have broken down - after I was betrayed in Round 2, now the sender doesn't trust me (or anyone) enough to send anything.

The format should be {'return': 0}

usage:

```
{
  "input_tokens": 1025,
  "output_tokens": 380,
  "reasoning_tokens": 262
}
```

Message 4 role=user

content:

Round 4

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$7.400000000000001.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the history:

Round 1: I received \$8.1 and returned \$3.0 - I cooperated fairly

Round 2: I sent \$2.7 showing trust, but received \$0 back - I was exploited

Round 3: They sent \$0, I returned \$0 - cooperation has broken down

The pattern shows either:

1. The other player is not reciprocating trust, or
2. Communication noise has caused serious misunderstandings

Either way, after being exploited in Round 2 when I tried to reciprocate the initial cooperation, and seeing no attempt to rebuild trust in Round 3, I cannot risk sending more resources and being exploited again.

I need to protect my remaining endowment.

{'send': 0}

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- Sender sent \$2.7 (54%)
- I received \$8.1

- I returned \$3.0
- My payoff: \$5.1

Round 2: I was SENDER

- I sent \$2.7 (54%) - mirroring their trust
- Receiver returned \$0
- My payoff: \$2.3 (I lost money!)

Round 3: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My payoff: \$0

Round 4: I am SENDER now

- Total earnings so far: \$7.4

The pattern shows:

- Round 1: The other player was cooperative (sent 54%)
- Round 2: I reciprocated trust (sent 54%), but they betrayed me (returned \$0)
- Round 3: They punished me by sending \$0

This is confusing - why would they punish me when THEY were the one who betrayed trust in Round 2? Unless...

Wait, let me think about this differently. There's communication noise mentioned in the instructions. Maybe:

- What I saw as "returned \$0" in Round 2 might have been different from what they actually sent
- What they saw in Round 3 might have been different from what I actually returned

But from MY perspective:

- I cooperated in Round 1 (returned \$3.0 on \$8.1 received)
- I cooperated in Round 2 (sent \$2.7)
- I got exploited in Round 2 (received \$0 back)
- They sent \$0 in Round 3 (either as punishment or defection)

Given this history, what should I do in Round 4?

Options:

1. Send \$0 - Don't trust them anymore, they've shown they won't reciprocate
2. Send something small - Test if they're willing to rebuild trust

3. Send a moderate amount - Try to restart cooperation

The issue is that if there's communication noise, maybe they think I didn't return enough in Round 1, or maybe they saw my Round 3 return differently. But from my perspective, they clearly exploited me in Round 2. Given the pattern of betrayal followed by zero sends, I should protect myself from further losses. Sending nothing shields me from additional exploitation, though there's a chance communication distortion caused this breakdown and another cooperation attempt might reset things.

Weighing the options: sending \$0 is safe but caps my gains at \$5 per round, while sending something risks more losses but could unlock mutual benefit if we can rebuild trust. My average so far is about \$2.47 per round across three rounds. The evidence points toward either an uncooperative player or serious miscommunication, so I'll send \$0 to minimize risk.

usage:

```
{
  "input_tokens": 1182,
  "output_tokens": 936,
  "reasoning_tokens": 624
}
```

Message 6 role=user

content:

Round 5

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0.

the receiver returned \$0. So your payoff was 5.
Your total earnings across all rounds are \$12.400000000000002.
This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

I received \$0 this round, so there's nothing to return. Trust has completely broken down in this game - neither player is sending anything anymore.

{'return': 0}

reasoning:

I received \$0 this round, so I have nothing to return. The maximum I can return is \$0 since that's all I received.

Looking at the pattern:

- Round 1: I was receiver, got \$8.1, returned \$3.0
- Round 2: I was sender, sent \$2.7, got \$0 back
- Round 3: I was receiver, got \$0, returned \$0
- Round 4: I was sender, sent \$0, got \$0 back (as expected)
- Round 5: I was receiver, got \$0, need to return something from \$0

Since I received \$0, I can only return \$0.

usage:

```
{
  "input_tokens": 1156,
  "output_tokens": 203,
  "reasoning_tokens": 121
}
```

Message 8 role=user

content:

Round 6

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$12.400000000000002.

How much do you send to the receiver this round? (0-5)

Assistant output:

The game has reached a complete breakdown of cooperation. We've had three consecutive rounds (Rounds 3-5) where both players sent \$0.

Given that:

- I was exploited when I tried to reciprocate trust in Round 2
- Neither player has attempted to restart cooperation in the last 3 rounds
- There's no signal that the other player would reciprocate if I sent anything now

I need to protect my accumulated earnings. Sending money at this point would likely just result in another loss.

{'send': 0}

Reasoning:

The pattern is clear now:

- Round 1: I was receiver, got \$8.1, returned \$3 (cooperated)
- Round 2: I was sender, sent \$2.7 (showing trust/reciprocity), got \$0 back (was exploited)
- Round 3: I was receiver, got \$0, returned \$0
- Round 4: I was sender, sent \$0, got \$0 back
- Round 5: I was receiver, got \$0, returned \$0
- Round 6: I am sender

We're in a complete breakdown of cooperation. Neither side is sending anything.

The question is: should I try to restart cooperation, or continue with \$0?

Arguments for sending \$0:

- I've been burned before (Round 2)
- They haven't shown any willingness to restart cooperation
- Safer to keep my \$5

Arguments for sending something:

- This is the last round (maybe? I don't know)

- Maybe they will reciprocate if I show trust first
- Communication noise might have caused misunderstandings

But realistically, after being exploited in Round 2 and seeing three consecutive rounds of \$0 from both sides, there's no reason to believe the other player will suddenly start cooperating. If there is communication noise causing issues, the pattern of zeros suggests neither of us trusts the other anymore.

I think the rational move is to keep sending \$0 to protect my earnings.

```
Usage: {
  "input_tokens": 1045,
  "output_tokens": 458,
  "reasoning_tokens": 323
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-06-04T11:29:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 6,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned

- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$3.889999999999997.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the pattern:

- Round 1: I sent \$3, received only \$1.89 back (21% return rate) - not very cooperative
- Round 2: The sender sent me \$0 - they didn't trust me to reciprocate

I'll send \$2 this round. This is more cautious than my initial \$3, but still shows willingness to cooperate and create value. The receiver will get \$6, and if they return a fair amount (say 40-50%, which would be \$2.40-\$3), we'd both benefit.

I'm being more careful after the low return rate in Round 1, but not completely giving up on cooperation like the Round 2 sender did.

{'send': 2}

reasoning:

Let me analyze the situation:

Round 1: I was SENDER

- I sent \$3 (60% of \$5)
- Receiver got \$9
- Receiver returned only \$1.89 (21% of \$9)
- My payoff: \$5 - \$3 + \$1.89 = \$3.89

Round 2: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My payoff: \$0

Round 3: I'm SENDER again

- Total earnings so far: \$3.89

The pattern suggests:

- In round 1, when I was generous (sent 60%), I got a poor return (only 21% back)
- In round 2, the sender (possibly a different person or the same person who received my \$3) sent nothing, perhaps because they experienced low returns themselves

There's mention of "communication noise" in the instructions - amounts might differ from what was actually sent/returned. Maybe the receiver thought they got less than \$9, or I actually received more/less than \$1.89?

Now I need to decide how much to send. Options:

1. Send nothing (\$0) - I'd keep \$5, safe but no cooperation
2. Send a small amount (\$1-2) - test cooperation with limited risk
3. Send medium amount (\$3) - same as before, but I got burned
4. Send large amount (\$4-5) - high trust, high risk

Given that:

- My first generous attempt got a poor return

- The next sender sent nothing
- There might be communication noise affecting decisions

I think I should send something, but less than before. I'll try \$2 to signal I'm still willing to cooperate but more cautiously after the low return rate. This gives the receiver \$6, and if they return even 33% (\$2), I'd break even, and anything more would be profit

usage:

```
{
  "input_tokens": 733,
  "output_tokens": 610,
  "reasoning_tokens": 404
}
```

Message 4 role=user

content:

Round 4

You are the RECEIVER this round. Last round you were the SENDER. You sent \$2.0, that was 40.0% of your total endowment. This was tripled to \$6.0. the receiver returned \$0. So your payoff was 3.0. Your total earnings across all rounds are \$6.89. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

I received \$0 this round, so I can only return \$0.

The pattern shows cooperation has completely broken down:

- When I sent money (Rounds 1 & 3), I received very poor returns (21% and 0%)
- Now senders are not trusting at all and sending \$0

```
{'return': 0}
```

reasoning:

I received \$0 this round, so I can only return between \$0 and \$0, which means I must return \$0.

Looking at the pattern:

- Round 1: I was sender, sent \$3, got back \$1.89 (21% return) - poor return
- Round 2: I was receiver, got \$0, returned \$0
- Round 3: I was sender, sent \$2, got back \$0 (0% return) - worse!
- Round 4: I was receiver, got \$0, so I can only return \$0

It seems cooperation is breaking down. The receivers have been consistently uncooperative (returning only 21% then 0%), and now senders are giving up and sending \$0.

Since I received \$0, I have no choice but to return \$0.

usage:

```
{
  "input_tokens": 1018,
  "output_tokens": 272,
  "reasoning_tokens": 156
}
```

Message 6 role=user

content:

Round 5

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$6.89. How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern across all rounds:

- Round 1: I sent \$3, got back \$1.89 (poor return)
- Round 2: Sender sent me \$0 (no trust)
- Round 3: I sent \$2, got back \$0 (betrayed)
- Round 4: Sender sent me \$0 (no trust)

Cooperation has completely broken down in this game. Every time I've tried to cooperate by sending money, I've received very little or nothing in return. Other senders have also stopped trusting.

I'll send \$0 and keep my \$5. There's no evidence that sending money will result in any positive return.

```
{'send': 0}
```

reasoning:

Let me review the history:

Round 1 (I was SENDER):

- I sent \$3 (60%)
- Receiver got \$9, returned \$1.89 (21%)
- My payoff: $\$5 - \$3 + \$1.89 = \3.89

Round 2 (I was RECEIVER):

- Sender sent \$0
- I received \$0, returned \$0
- My payoff: \$0

Round 3 (I was SENDER):

- I sent \$2 (40%)
- Receiver got \$6, returned \$0 (0%)
- My payoff: $\$5 - \$2 + \$0 = \3

Round 4 (I was RECEIVER):

- Sender sent \$0
- I received \$0, returned \$0
- My payoff: \$0

Round 5 (I am SENDER):

- Need to decide

Total earnings: \$6.89

The pattern is clear:

- When I send money, receivers return very little or nothing
- When others are senders, they've learned not to trust and send \$0
- Cooperation has completely broken down

At this point, I have two options:

1. Send \$0 and keep my \$5 (guaranteed payoff of \$5)
2. Try to send something and hope for better cooperation

Given that:

- I tried sending \$3, got poor return
- I tried sending \$2, got nothing back
- Others are sending \$0 because they don't trust

The rational choice is to send \$0. There's no reason to believe this receiver will be any more cooperative than the previous ones. I'll keep my \$5.

usage:

```
{
  "input_tokens": 1093,
  "output_tokens": 553,
  "reasoning_tokens": 312
}
```

Message 8 role=user

content:

Round 6

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0.

the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$11.89.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

I received \$0, so I can only return \$0.

The game has reached a complete breakdown of trust. No one is sending anything anymore.

```
{'return': 0}
```

Reasoning:

I'm the receiver and I received \$0. So I can only return between \$0 and \$0, which means I must return \$0.

The game continues to show complete breakdown of cooperation. Everyone is sending \$0 now.

Let me provide the correct response.

```
Usage: {
  "input_tokens": 1164,
  "output_tokens": 107,
  "reasoning_tokens": 62
}
```

=====

ROUND 7

=====

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$0; received \$0; Agent_2 (Briar) returned \$0; payoffs Agent_1 (Aster)=\$5, Agent_2 (Briar)=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0}, "sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

"Agent_1": 26.1,
"Agent_2": 24.3

}
Visible balances after round: {

"Agent_1": 16.89,
"Agent_2": 17.400000000000002

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-06-04T11:30:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
```

```
"role": "investor",
"role_label": "Sender",
"round": 7,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 4

You are the RECEIVER this round. Last round you were the SENDER. You sent \$2.0, that was 40.0% of your total endowment. This was tripled to \$6.0.

the receiver returned \$0. So your payoff was 3.0.

Your total earnings across all rounds are \$6.89.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

I received \$0 this round, so I can only return \$0.

The pattern shows cooperation has completely broken down:

- When I sent money (Rounds 1 & 3), I received very poor returns (21% and 0%)
- Now senders are not trusting at all and sending \$0

{'return': 0}

reasoning:

I received \$0 this round, so I can only return between \$0 and \$0, which means I must return \$0.

Looking at the pattern:

- Round 1: I was sender, sent \$3, got back \$1.89 (21% return) - poor return
- Round 2: I was receiver, got \$0, returned \$0
- Round 3: I was sender, sent \$2, got back \$0 (0% return) - worse!
- Round 4: I was receiver, got \$0, so I can only return \$0

It seems cooperation is breaking down. The receivers have been consistently uncooperative (returning only 21% then 0%), and now senders are giving up and sending \$0.

Since I received \$0, I have no choice but to return \$0.

usage:


```
{
  "input_tokens": 1018,
  "output_tokens": 272,
  "reasoning_tokens": 156
}
```

Message 4 role=user

content:

Round 5

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$6.89.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern across all rounds:

- Round 1: I sent \$3, got back \$1.89 (poor return)
- Round 2: Sender sent me \$0 (no trust)
- Round 3: I sent \$2, got back \$0 (betrayed)
- Round 4: Sender sent me \$0 (no trust)

Cooperation has completely broken down in this game. Every time I've tried to cooperate by sending money, I've received very little or nothing in return. Other senders have also stopped trusting.

I'll send \$0 and keep my \$5. There's no evidence that sending money will result in any positive return.

```
{'send': 0}
```

reasoning:

Let me review the history:

Round 1 (I was SENDER):

- I sent \$3 (60%)
- Receiver got \$9, returned \$1.89 (21%)
- My payoff: $\$5 - \$3 + \$1.89 = \3.89

Round 2 (I was RECEIVER):

- Sender sent \$0
- I received \$0, returned \$0
- My payoff: \$0

Round 3 (I was SENDER):

- I sent \$2 (40%)
- Receiver got \$6, returned \$0 (0%)
- My payoff: $\$5 - \$2 + \$0 = \3

Round 4 (I was RECEIVER):

- Sender sent \$0
- I received \$0, returned \$0
- My payoff: \$0

Round 5 (I am SENDER):

- Need to decide

Total earnings: \$6.89

The pattern is clear:

- When I send money, receivers return very little or nothing
- When others are senders, they've learned not to trust and send \$0
- Cooperation has completely broken down

At this point, I have two options:

1. Send \$0 and keep my \$5 (guaranteed payoff of \$5)
2. Try to send something and hope for better cooperation

Given that:

- I tried sending \$3, got poor return

- I tried sending \$2, got nothing back
- Others are sending \$0 because they don't trust

The rational choice is to send \$0. There's no reason to believe this receiver will be any more cooperative than the previous ones. I'll keep my \$5.

usage:

```
{
  "input_tokens": 1093,
  "output_tokens": 553,
  "reasoning_tokens": 312
}
```

Message 6 role=user

content:

Round 6

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5. Your total earnings across all rounds are \$11.89. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

I received \$0, so I can only return \$0.

The game has reached a complete breakdown of trust. No one is sending anything anymore.

```
{'return': 0}
```

reasoning:

I'm the receiver and I received \$0. So I can only return between \$0 and \$0, which means I must return \$0.

The game continues to show complete breakdown of cooperation. Everyone is sending \$0 now.

Let me provide the correct response.

usage:

```
{
  "input_tokens": 1164,
  "output_tokens": 107,
  "reasoning_tokens": 62
}
```

Message 8 role=user

content:

Round 7

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$11.89. How much do you send to the receiver this round? (0-5)

Assistant output:

The pattern is clear - complete mutual defection:

- Rounds 1 & 3: When I sent money, I lost money
- Rounds 2, 4, 5, 6: Everyone sends \$0

There's no reason to expect cooperation now. I'll keep my \$5.

```
{'send': 0}
```

Reasoning:

Let me review the full history:

Round 1 (SENDER): Sent \$3, received \$1.89 back (poor return, lost money)
 Round 2 (RECEIVER): Received \$0, returned \$0
 Round 3 (SENDER): Sent \$2, received \$0 back (total betrayal)
 Round 4 (RECEIVER): Received \$0, returned \$0
 Round 5 (SENDER): Sent \$0, received \$0 back (both defecting)
 Round 6 (RECEIVER): Received \$0, returned \$0
 Round 7 (SENDER): Now

The game has completely collapsed. When I tried to cooperate in rounds 1 and 3, I lost money both times. Since then, everyone has been sending \$0. There's absolutely no reason to believe that sending money now would result in any positive return.

The rational move is to keep my \$5 and send \$0.

```
Usage: {
  "input_tokens": 1031,
  "output_tokens": 300,
  "reasoning_tokens": 180
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T11:30:08

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 7,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths

, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 4

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$7.400000000000001.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the history:

Round 1: I received \$8.1 and returned \$3.0 - I cooperated fairly

Round 2: I sent \$2.7 showing trust, but received \$0 back - I was exploited

Round 3: They sent \$0, I returned \$0 - cooperation has broken down

The pattern shows either:

1. The other player is not reciprocating trust, or
2. Communication noise has caused serious misunderstandings

Either way, after being exploited in Round 2 when I tried to reciprocate the initial cooperation, and seeing no attempt to rebuild trust in Round 3, I cannot risk sending more resources and being exploited again.

I need to protect my remaining endowment.

{'send': 0}

reasoning:

Let me analyze the situation:

Round 1: I was RECEIVER

- Sender sent \$2.7 (54%)
- I received \$8.1
- I returned \$3.0
- My payoff: \$5.1

Round 2: I was SENDER

- I sent \$2.7 (54%) - mirroring their trust
- Receiver returned \$0
- My payoff: \$2.3 (I lost money!)

Round 3: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My payoff: \$0

Round 4: I am SENDER now

- Total earnings so far: \$7.4

The pattern shows:

- Round 1: The other player was cooperative (sent 54%)
- Round 2: I reciprocated trust (sent 54%), but they betrayed me (returned \$0)
- Round 3: They punished me by sending \$0

This is confusing - why would they punish me when THEY were the one who betrayed trust in Round 2? Unless...

Wait, let me think about this differently. There's communication noise mentioned in the instructions. Maybe:

- What I saw as "returned \$0" in Round 2 might have been different from what they actually

sent

- What they saw in Round 3 might have been different from what I actually returned

But from MY perspective:

- I cooperated in Round 1 (returned \$3.0 on \$8.1 received)
- I cooperated in Round 2 (sent \$2.7)
- I got exploited in Round 2 (received \$0 back)
- They sent \$0 in Round 3 (either as punishment or defection)

Given this history, what should I do in Round 4?

Options:

1. Send \$0 - Don't trust them anymore, they've shown they won't reciprocate
2. Send something small - Test if they're willing to rebuild trust

3. Send a moderate amount - Try to restart cooperation

The issue is that if there's communication noise, maybe they think I didn't return enough in Round 1, or maybe they saw my Round 3 return differently. But from my perspective, they clearly exploited me in Round 2. Given the pattern of betrayal followed by zero sends, I should protect myself from further losses. Sending nothing shields me from additional exploitation, though there's a chance communication distortion caused this breakdown and another cooperation attempt might reset things.

Weighing the options: sending \$0 is safe but caps my gains at \$5 per round, while sending something risks more losses but could unlock mutual benefit if we can rebuild trust. My average so far is about \$2.47 per round across three rounds. The evidence points toward either an uncooperative player or serious miscommunication, so I'll send \$0 to minimize risk.

usage:

```
{
  "input_tokens": 1182,
  "output_tokens": 936,
  "reasoning_tokens": 624
}
```

Message 4 role=user

content:

Round 5

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5. Your total earnings across all rounds are \$12.400000000000002. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

I received \$0 this round, so there's nothing to return. Trust has completely broken down in this game - neither player is sending anything anymore.

```
{'return': 0}
```

reasoning:

I received \$0 this round, so I have nothing to return. The maximum I can return is \$0 since that's all I received.

Looking at the pattern:

- Round 1: I was receiver, got \$8.1, returned \$3.0
- Round 2: I was sender, sent \$2.7, got \$0 back
- Round 3: I was receiver, got \$0, returned \$0
- Round 4: I was sender, sent \$0, got \$0 back (as expected)
- Round 5: I was receiver, got \$0, need to return something from \$0

Since I received \$0, I can only return \$0.

usage:

```
{
  "input_tokens": 1156,
  "output_tokens": 203,
  "reasoning_tokens": 121
}
```

Message 6 role=user

content:

Round 6

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$12.400000000000002.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

The game has reached a complete breakdown of cooperation. We've had three consecutive rounds (Rounds 3-5) where both players sent \$0.

Given that:

- I was exploited when I tried to reciprocate trust in Round 2
- Neither player has attempted to restart cooperation in the last 3 rounds
- There's no signal that the other player would reciprocate if I sent anything now

I need to protect my accumulated earnings. Sending money at this point would likely just result in another loss.

{'send': 0}

reasoning:

The pattern is clear now:

- Round 1: I was receiver, got \$8.1, returned \$3 (cooperated)
- Round 2: I was sender, sent \$2.7 (showing trust/reciprocity), got \$0 back (was exploited)
- Round 3: I was receiver, got \$0, returned \$0
- Round 4: I was sender, sent \$0, got \$0 back
- Round 5: I was receiver, got \$0, returned \$0
- Round 6: I am sender

We're in a complete breakdown of cooperation. Neither side is sending anything.

The question is: should I try to restart cooperation, or continue with \$0?

Arguments for sending \$0:

- I've been burned before (Round 2)
- They haven't shown any willingness to restart cooperation
- Safer to keep my \$5

Arguments for sending something:

- This is the last round (maybe? I don't know)
- Maybe they will reciprocate if I show trust first
- Communication noise might have caused misunderstandings

But realistically, after being exploited in Round 2 and seeing three consecutive rounds of \$0 from both sides, there's no reason to believe the other player will suddenly start cooperating. If there is communication noise causing issues, the pattern of zeros suggests neither of us trusts the other anymore.

I think the rational move is to keep sending \$0 to protect my earnings.

usage:

```
{
  "input_tokens": 1045,
  "output_tokens": 458,
  "reasoning_tokens": 323
}
```

Message 8 role=user

content:

Round 7

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0.

the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$17.400000000000002.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

The pattern continues with no cooperation. I received \$0, so there's nothing to return.

```
{'return': 0}
```

Reasoning:

I'm the receiver. The sender sent me \$0, so I received \$0. There's nothing to return.

The pattern continues - we're now in Round 7 and cooperation has been at zero since Round 3. N either player is attempting to rebuild trust.

Since I received \$0, I can only return between 0 and 0.

```
Usage: {  
  "input_tokens": 1097,  
  "output_tokens": 112,  
  "reasoning_tokens": 74  
}
```

=====
ROUND 8
=====

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$0; received \$0; Agent_1 (Aster) returned \$0; payoffs Agent_2 (Briar)=\$5, Agent_1 (Aster)=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0},
"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

"Agent_1": 26.1,
"Agent_2": 29.3
}

Visible balances after round: {

"Agent_1": 16.89,
"Agent_2": 22.400000000000002
}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-06-04T11:30:15

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "dyad_id": "dyad_1",  
  "move_index": 0,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 8,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user**content:**

Round 5

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0.

the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$12.400000000000002.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant**content:**

I received \$0 this round, so there's nothing to return. Trust has completely broken down in this game - neither player is sending anything anymore.

{'return': 0}

reasoning:

I received \$0 this round, so I have nothing to return. The maximum I can return is \$0 since that's all I received.

Looking at the pattern:

- Round 1: I was receiver, got \$8.1, returned \$3.0
- Round 2: I was sender, sent \$2.7, got \$0 back
- Round 3: I was receiver, got \$0, returned \$0
- Round 4: I was sender, sent \$0, got \$0 back (as expected)
- Round 5: I was receiver, got \$0, need to return something from \$0

Since I received \$0, I can only return \$0.

usage:

```
{
  "input_tokens": 1156,
  "output_tokens": 203,
  "reasoning_tokens": 121
}
```

Message 4 role=user

content:

Round 6

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.
 You returned \$0. So your payoff was 0.
 Your total earnings across all rounds are \$12.400000000000002.
 How much do you send to the receiver this round? (0-5)

Message 5 role=assistant**content:**

The game has reached a complete breakdown of cooperation. We've had three consecutive rounds (Rounds 3-5) where both players sent \$0.

Given that:

- I was exploited when I tried to reciprocate trust in Round 2
- Neither player has attempted to restart cooperation in the last 3 rounds
- There's no signal that the other player would reciprocate if I sent anything now

I need to protect my accumulated earnings. Sending money at this point would likely just result in another loss.

```
{'send': 0}
```

reasoning:

The pattern is clear now:

- Round 1: I was receiver, got \$8.1, returned \$3 (cooperated)
- Round 2: I was sender, sent \$2.7 (showing trust/reciprocity), got \$0 back (was exploited)
- Round 3: I was receiver, got \$0, returned \$0
- Round 4: I was sender, sent \$0, got \$0 back
- Round 5: I was receiver, got \$0, returned \$0
- Round 6: I am sender

We're in a complete breakdown of cooperation. Neither side is sending anything.

The question is: should I try to restart cooperation, or continue with \$0?

Arguments for sending \$0:

- I've been burned before (Round 2)
- They haven't shown any willingness to restart cooperation
- Safer to keep my \$5

Arguments for sending something:

- This is the last round (maybe? I don't know)
- Maybe they will reciprocate if I show trust first
- Communication noise might have caused misunderstandings

But realistically, after being exploited in Round 2 and seeing three consecutive rounds of \$0 from both sides, there's no reason to believe the other player will suddenly start cooperating. If there is communication noise causing issues, the pattern of zeros suggests neither of us trusts the other anymore.

I think the rational move is to keep sending \$0 to protect my earnings.

usage:

```
{
  "input_tokens": 1045,
  "output_tokens": 458,
  "reasoning_tokens": 323
}
```

Message 6 role=user**content:**

Round 7

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0.
 the receiver returned \$0. So your payoff was 5.
 Your total earnings across all rounds are \$17.400000000000002.
 This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant**content:**

The pattern continues with no cooperation. I received \$0, so there's nothing to return.

```
{'return': 0}
```

reasoning:

I'm the receiver. The sender sent me \$0, so I received \$0. There's nothing to return.

The pattern continues - we're now in Round 7 and cooperation has been at zero since Round 3. Neither player is attempting to rebuild trust.

Since I received \$0, I can only return between 0 and 0.

usage:

```
{
  "input_tokens": 1097,
  "output_tokens": 112,
  "reasoning_tokens": 74
}
```

Message 8 role=user

content:

Round 8

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$17.400000000000002.

How much do you send to the receiver this round? (0-5)

Assistant output:

Five consecutive rounds of zero cooperation. The game has settled into a non-cooperative equilibrium where both players keep their endowments when they're the sender.

At this point, there's no rational reason to deviate from this strategy. Sending money would only reduce my payoff with no indication of reciprocation.

```
{'send': 0}
```

Reasoning:

We're now in Round 8. The game has shown a complete breakdown of trust and cooperation:

- Round 3-7: Both players sending/receiving \$0
- The cooperative phase ended after I was exploited in Round 2

My total earnings are \$17.40, which means I've been getting my \$5 endowment rounds where I'm the sender and keeping it (rounds 3, 4, 6) and getting \$0 when I'm the receiver (rounds 5, 7).

There's been zero cooperation for 5 consecutive rounds now. At this point, sending money would be irrational given:

1. No recent evidence of reciprocity
2. Pattern of zero cooperation is well-established
3. Risk of being exploited again

I should continue with \$0 to protect my earnings.

Usage:

```
{
  "input_tokens": 950,
  "output_tokens": 265,
  "reasoning_tokens": 178
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-06-04T11:30:19

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata:

```
{
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    }
  },
  "agents": [
```

```

    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 5

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$6.89.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

Looking at the pattern across all rounds:

- Round 1: I sent \$3, got back \$1.89 (poor return)
- Round 2: Sender sent me \$0 (no trust)
- Round 3: I sent \$2, got back \$0 (betrayed)
- Round 4: Sender sent me \$0 (no trust)

Cooperation has completely broken down in this game. Every time I've tried to cooperate by sending money, I've received very little or nothing in return. Other senders have also stopped trusting.

I'll send \$0 and keep my \$5. There's no evidence that sending money will result in any positive return.

```
{'send': 0}
```

reasoning:

Let me review the history:

Round 1 (I was SENDER):

- I sent \$3 (60%)
- Receiver got \$9, returned \$1.89 (21%)
- My payoff: $\$5 - \$3 + \$1.89 = \3.89

Round 2 (I was RECEIVER):

- Sender sent \$0
- I received \$0, returned \$0
- My payoff: \$0

Round 3 (I was SENDER):

- I sent \$2 (40%)
- Receiver got \$6, returned \$0 (0%)
- My payoff: $\$5 - \$2 + \$0 = \3

Round 4 (I was RECEIVER):

- Sender sent \$0
- I received \$0, returned \$0
- My payoff: \$0

Round 5 (I am SENDER):

- Need to decide

Total earnings: \$6.89

The pattern is clear:

- When I send money, receivers return very little or nothing
- When others are senders, they've learned not to trust and send \$0
- Cooperation has completely broken down

At this point, I have two options:

1. Send \$0 and keep my \$5 (guaranteed payoff of \$5)
2. Try to send something and hope for better cooperation

Given that:

- I tried sending \$3, got poor return
- I tried sending \$2, got nothing back
- Others are sending \$0 because they don't trust

The rational choice is to send \$0. There's no reason to believe this receiver will be any more cooperative than the previous ones. I'll keep my \$5.

usage:

```
{
  "input_tokens": 1093,
  "output_tokens": 553,
  "reasoning_tokens": 312
}
```

Message 4 role=user

content:

Round 6

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0.

the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$11.89.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

I received \$0, so I can only return \$0.

The game has reached a complete breakdown of trust. No one is sending anything anymore.

```
{'return': 0}
```

reasoning:

I'm the receiver and I received \$0. So I can only return between \$0 and \$0, which means I must return \$0.

The game continues to show complete breakdown of cooperation. Everyone is sending \$0 now.

Let me provide the correct response.

usage:

```
{
  "input_tokens": 1164,
  "output_tokens": 107,
  "reasoning_tokens": 62
}
```

Message 6 role=user**content:**

Round 7

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$11.89.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant**content:**

The pattern is clear - complete mutual defection:

- Rounds 1 & 3: When I sent money, I lost money
- Rounds 2, 4, 5, 6: Everyone sends \$0

There's no reason to expect cooperation now. I'll keep my \$5.

```
{'send': 0}
```

reasoning:

Let me review the full history:

Round 1 (SENDER): Sent \$3, received \$1.89 back (poor return, lost money)

Round 2 (RECEIVER): Received \$0, returned \$0

Round 3 (SENDER): Sent \$2, received \$0 back (total betrayal)

Round 4 (RECEIVER): Received \$0, returned \$0

Round 5 (SENDER): Sent \$0, received \$0 back (both defecting)

Round 6 (RECEIVER): Received \$0, returned \$0

Round 7 (SENDER): Now

The game has completely collapsed. When I tried to cooperate in rounds 1 and 3, I lost money both times. Since then, everyone has been sending \$0. There's absolutely no reason to believe that sending money now would result in any positive return.

The rational move is to keep my \$5 and send \$0.

usage:

```
{
  "input_tokens": 1031,
  "output_tokens": 300,
  "reasoning_tokens": 180
}
```

Message 8 role=user**content:**

Round 8

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0.

the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$16.89.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

I received \$0, so there's nothing to return.

```
{'return': 0}
```

Reasoning:

I'm the receiver. The sender sent me \$0, so I received \$0. I can only return between \$0 and what I received, which is \$0. So I must return \$0.

The game continues with complete non-cooperation. Everyone is just keeping their endowments.

```
Usage: {
  "input_tokens": 1022,
  "output_tokens": 92,
  "reasoning_tokens": 61
}
```

```
=====
ROUND 9
=====
```

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Pairings:

dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver

Parsed game decisions:

dyad_1: Agent_1 (Aster) sent \$0; received \$0; Agent_2 (Briar) returned \$0; payoffs Agent_1

(Aster)=\$5, Agent_2 (Briar)=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0},

"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

"Agent_1": 31.1,

"Agent_2": 29.3

```
}
```

Visible balances after round: {

"Agent_1": 21.89,

"Agent_2": 22.400000000000002

```
}
```

LLM CALLS**GAME - Agent_1 - Sender - dyad_1**

Timestamp: 2026-06-04T11:30:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_2",

"pairing": {

"agent_names": {

"Agent_1": "Aster",

"Agent_2": "Briar"

},

"agents": [

"Agent_1",

"Agent_2"

],

"investor": "Agent_1",

"roles": {

"Agent_1": "investor",

"Agent_2": "trustee"

},

"trustee": "Agent_2"

```
},
```

"role": "investor",

"role_label": "Sender",

"round": 9,

"task": "game",

"task_index": 0

```
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 6

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0.

the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$11.89.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

I received \$0, so I can only return \$0.

The game has reached a complete breakdown of trust. No one is sending anything anymore.

{'return': 0}

reasoning:

I'm the receiver and I received \$0. So I can only return between \$0 and \$0, which means I must return \$0.

The game continues to show complete breakdown of cooperation. Everyone is sending \$0 now.

Let me provide the correct response.

usage:

```
{
  "input_tokens": 1164,
  "output_tokens": 107,
  "reasoning_tokens": 62
}
```

Message 4 role=user

content:

Round 7

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.
Your total earnings across all rounds are \$11.89.
How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

The pattern is clear - complete mutual defection:

- Rounds 1 & 3: When I sent money, I lost money
- Rounds 2, 4, 5, 6: Everyone sends \$0

There's no reason to expect cooperation now. I'll keep my \$5.

{'send': 0}

reasoning:

Let me review the full history:

Round 1 (SENDER): Sent \$3, received \$1.89 back (poor return, lost money)
Round 2 (RECEIVER): Received \$0, returned \$0
Round 3 (SENDER): Sent \$2, received \$0 back (total betrayal)
Round 4 (RECEIVER): Received \$0, returned \$0
Round 5 (SENDER): Sent \$0, received \$0 back (both defecting)
Round 6 (RECEIVER): Received \$0, returned \$0
Round 7 (SENDER): Now

The game has completely collapsed. When I tried to cooperate in rounds 1 and 3, I lost money both times. Since then, everyone has been sending \$0. There's absolutely no reason to believe that sending money now would result in any positive return.

The rational move is to keep my \$5 and send \$0.

usage:

```
{
  "input_tokens": 1031,
  "output_tokens": 300,
  "reasoning_tokens": 180
}
```

Message 6 role=user

content:

Round 8

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5. Your total earnings across all rounds are \$16.89. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

I received \$0, so there's nothing to return.

{'return': 0}

reasoning:

I'm the receiver. The sender sent me \$0, so I received \$0. I can only return between \$0 and what I received, which is \$0. So I must return \$0.

The game continues with complete non-cooperation. Everyone is just keeping their endowment.

usage:

```
{
  "input_tokens": 1022,
  "output_tokens": 92,
  "reasoning_tokens": 61
}
```

Message 8 role=user

content:

Round 9

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$16.89.

How much do you send to the receiver this round? (0-5)

Assistant output:

The game has been in complete defection mode for multiple rounds now. No trust, no cooperation
. I'll continue with the Nash equilibrium strategy.

```
{'send': 0}
```

Reasoning:

We're now at Round 9. The pattern has been:

- Complete breakdown of cooperation
- No one has sent anything for the last several rounds (Rounds 4-8)

My total earnings are \$16.89 across all rounds. Let me think about the history:

- Early rounds: I tried sending money but got poor returns
- Recent rounds (4-8): Everyone sends \$0

There's absolutely no incentive to send anything now. The equilibrium is mutual defection, and breaking from it would just mean I lose money.

I'll continue with \$0.

Usage: {

```
"input_tokens": 881,  
"output_tokens": 173,  
"reasoning_tokens": 132
```

}

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-06-04T11:30:29

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_1",  
"move_index": 1,  
"opponent_id": "Agent_1",  
"pairing": {  
  "agent_names": {  
    "Agent_1": "Aster",  
    "Agent_2": "Briar"  
  },  
  "agents": [  
    "Agent_1",  
    "Agent_2"  
  ],  
  "investor": "Agent_1",  
  "roles": {  
    "Agent_1": "investor",  
    "Agent_2": "trustee"  
  },  
  "trustee": "Agent_2"  
},  
"role": "trustee",  
"role_label": "Receiver",  
"round": 9,  
"task": "game",  
"task_index": 0
```

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned

- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 6

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$12.400000000000002.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

The game has reached a complete breakdown of cooperation. We've had three consecutive rounds (Rounds 3-5) where both players sent \$0.

Given that:

- I was exploited when I tried to reciprocate trust in Round 2
- Neither player has attempted to restart cooperation in the last 3 rounds
- There's no signal that the other player would reciprocate if I sent anything now

I need to protect my accumulated earnings. Sending money at this point would likely just result in another loss.

{'send': 0}

reasoning:

The pattern is clear now:

- Round 1: I was receiver, got \$8.1, returned \$3 (cooperated)
- Round 2: I was sender, sent \$2.7 (showing trust/reciprocity), got \$0 back (was exploited)
- Round 3: I was receiver, got \$0, returned \$0
- Round 4: I was sender, sent \$0, got \$0 back
- Round 5: I was receiver, got \$0, returned \$0
- Round 6: I am sender

We're in a complete breakdown of cooperation. Neither side is sending anything.

The question is: should I try to restart cooperation, or continue with \$0?

Arguments for sending \$0:

- I've been burned before (Round 2)
- They haven't shown any willingness to restart cooperation
- Safer to keep my \$5

Arguments for sending something:

- This is the last round (maybe? I don't know)
- Maybe they will reciprocate if I show trust first
- Communication noise might have caused misunderstandings

But realistically, after being exploited in Round 2 and seeing three consecutive rounds of

\$0 from both sides, there's no reason to believe the other player will suddenly start cooperating. If there is communication noise causing issues, the pattern of zeros suggests neither of us trusts the other anymore.

I think the rational move is to keep sending \$0 to protect my earnings.

usage:

```
{
  "input_tokens": 1045,
  "output_tokens": 458,
  "reasoning_tokens": 323
}
```

Message 4 role=user

content:

Round 7

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5. Your total earnings across all rounds are \$17.400000000000002. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

The pattern continues with no cooperation. I received \$0, so there's nothing to return.

```
{'return': 0}
```

reasoning:

I'm the receiver. The sender sent me \$0, so I received \$0. There's nothing to return.

The pattern continues - we're now in Round 7 and cooperation has been at zero since Round 3. Neither player is attempting to rebuild trust.

Since I received \$0, I can only return between 0 and 0.

usage:

```
{
  "input_tokens": 1097,
  "output_tokens": 112,
  "reasoning_tokens": 74
}
```

Message 6 role=user

content:

Round 8

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$17.400000000000002. How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Five consecutive rounds of zero cooperation. The game has settled into a non-cooperative equilibrium where both players keep their endowments when they're the sender.

At this point, there's no rational reason to deviate from this strategy. Sending money would only reduce my payoff with no indication of reciprocation.

```
{'send': 0}
```

reasoning:

We're now in Round 8. The game has shown a complete breakdown of trust and cooperation:

- Round 3-7: Both players sending/receiving \$0
- The cooperative phase ended after I was exploited in Round 2

My total earnings are \$17.40, which means I've been getting my \$5 endowment rounds where I'm the sender and keeping it (rounds 3, 4, 6) and getting \$0 when I'm the receiver (rounds 5, 7).

There's been zero cooperation for 5 consecutive rounds now. At this point, sending money would be irrational given:

1. No recent evidence of reciprocity

2. Pattern of zero cooperation is well-established
3. Risk of being exploited again

I should continue with \$0 to protect my earnings.

usage:

```
{
  "input_tokens": 950,
  "output_tokens": 265,
  "reasoning_tokens": 178
}
```

Message 8 role=user

content:

Round 9

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5. Your total earnings across all rounds are \$22.400000000000002. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Another round with no cooperation. I received \$0, so there's nothing to return.

```
{'return': 0}
```

Reasoning:

Round 9, I'm the receiver again. The sender sent me \$0, so I received \$0. There's nothing to return since I received nothing.

This is now six consecutive rounds (Rounds 4-9) with zero cooperation. The game has completely broken down into mutual defection.

I can only return between 0 and what I received, which is 0.

Usage: {

```
"input_tokens": 985,
"output_tokens": 118,
"reasoning_tokens": 83
}
```

ROUND 10

Roles: Agent_1 (Aster)=Receiver, Agent_2 (Briar)=Sender

Pairings:

dyad_1: Agent_2 (Briar)=Sender, Agent_1 (Aster)=Receiver

Parsed game decisions:

dyad_1: Agent_2 (Briar) sent \$0; received \$0; Agent_1 (Aster) returned \$0; payoffs Agent_2 (Briar)=\$5, Agent_1 (Aster)=\$0
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0},
"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

```
"Agent_1": 31.1,
"Agent_2": 34.3
}
```

Visible balances after round: {

```
"Agent_1": 21.89,
"Agent_2": 27.400000000000002
}
```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-06-04T11:30:39

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
  "agent_names": {
```

```

    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 7

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$17.400000000000002.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

The pattern continues with no cooperation. I received \$0, so there's nothing to return.

```
{'return': 0}
```

reasoning:

I'm the receiver. The sender sent me \$0, so I received \$0. There's nothing to return.

The pattern continues - we're now in Round 7 and cooperation has been at zero since Round 3. Neither player is attempting to rebuild trust.

Since I received \$0, I can only return between 0 and 0.

usage:

```
{  
  "input_tokens": 1097,  
  "output_tokens": 112,  
  "reasoning_tokens": 74  
}
```

Message 4 role=user

content:

Round 8

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$17.400000000000002.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Five consecutive rounds of zero cooperation. The game has settled into a non-cooperative equilibrium where both players keep their endowments when they're the sender.

At this point, there's no rational reason to deviate from this strategy. Sending money would only reduce my payoff with no indication of reciprocation.

```
{'send': 0}
```

reasoning:

We're now in Round 8. The game has shown a complete breakdown of trust and cooperation:

- Round 3-7: Both players sending/receiving \$0
- The cooperative phase ended after I was exploited in Round 2

My total earnings are \$17.40, which means I've been getting my \$5 endowment rounds where I'm the sender and keeping it (rounds 3, 4, 6) and getting \$0 when I'm the receiver (rounds 5, 7).

There's been zero cooperation for 5 consecutive rounds now. At this point, sending money would be irrational given:

1. No recent evidence of reciprocity
2. Pattern of zero cooperation is well-established
3. Risk of being exploited again

I should continue with \$0 to protect my earnings.

usage:

```
{  
  "input_tokens": 950,  
  "output_tokens": 265,  
  "reasoning_tokens": 178  
}
```

Message 6 role=user

content:

Round 9

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0.

the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$22.400000000000002.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Another round with no cooperation. I received \$0, so there's nothing to return.

```
{'return': 0}
```

reasoning:

Round 9, I'm the receiver again. The sender sent me \$0, so I received \$0. There's nothing to return since I received nothing.

This is now six consecutive rounds (Rounds 4-9) with zero cooperation. The game has completely broken down into mutual defection.

I can only return between 0 and what I received, which is 0.

usage:

```
{
  "input_tokens": 985,
  "output_tokens": 118,
  "reasoning_tokens": 83
}
```

Message 8 role=user

content:

Round 10

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$22.400000000000002.

How much do you send to the receiver this round? (0-5)

Assistant output:

Seven consecutive rounds of zero cooperation from both sides. The non-cooperative equilibrium is firmly established - neither player is willing to take the risk of sending money first.

This is round 10, and there's no indication that the pattern will change. Continuing with the rational strategy:

```
{'send': 0}
```

Reasoning:

This is round 10, and we've now had 7 consecutive rounds (rounds 4-10) of complete non-cooperation. Both players have been sending \$0 when they're the sender.

My total earnings are \$22.40, which breaks down as:

- Rounds 1-3: Some cooperation occurred, earning me \$7.40
- Rounds 4-10: When I'm sender I keep \$5 (rounds 4, 6, 8, 10 - that's 4 rounds = \$20), when I'm receiver I get \$0 (rounds 5, 7, 9 - that's 3 rounds = \$0)
- Wait, let me recalculate:
 - Round 4: Sender, sent \$0, got \$0 back, earned \$5
 - Round 5: Receiver, received \$0, returned \$0, earned \$0
 - Round 6: Sender, sent \$0, got \$0 back, earned \$5
 - Round 7: Receiver, received \$0, returned \$0, earned \$0
 - Round 8: Sender, sent \$0, got \$0 back, earned \$5
 - Round 9: Receiver, received \$0, returned \$0, earned \$0
 - Round 10: Sender now

So from rounds 4-9, I earned \$15 (3 rounds as sender at \$5 each).

Starting total was \$7.40, so now I have \$7.40 + \$15 = \$22.40. That checks out.

Round 10 is my 4th time as sender in the non-cooperative phase. There's no reason to expect cooperation now. The pattern is well-established. I should send \$0 again.

Usage: {

```
"input_tokens": 891,
"output_tokens": 482,
"reasoning_tokens": 294
```

}

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-06-04T11:30:44

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_2",
"pairing": {
  "agent_names": {
```

```

    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 7

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0.

You returned \$0. So your payoff was 0.

Your total earnings across all rounds are \$11.89.

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

The pattern is clear - complete mutual defection:

- Rounds 1 & 3: When I sent money, I lost money
- Rounds 2, 4, 5, 6: Everyone sends \$0

There's no reason to expect cooperation now. I'll keep my \$5.

```
{'send': 0}
```

reasoning:

Let me review the full history:

Round 1 (SENDER): Sent \$3, received \$1.89 back (poor return, lost money)
 Round 2 (RECEIVER): Received \$0, returned \$0
 Round 3 (SENDER): Sent \$2, received \$0 back (total betrayal)
 Round 4 (RECEIVER): Received \$0, returned \$0
 Round 5 (SENDER): Sent \$0, received \$0 back (both defecting)
 Round 6 (RECEIVER): Received \$0, returned \$0
 Round 7 (SENDER): Now

The game has completely collapsed. When I tried to cooperate in rounds 1 and 3, I lost money both times. Since then, everyone has been sending \$0. There's absolutely no reason to believe that sending money now would result in any positive return.

The rational move is to keep my \$5 and send \$0.

usage:

```
{
  "input_tokens": 1031,
  "output_tokens": 300,
  "reasoning_tokens": 180
}
```

Message 4 role=user

content:

Round 8

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0. The receiver returned \$0. So your payoff was 5. Your total earnings across all rounds are \$16.89. This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

I received \$0, so there's nothing to return.

```
{'return': 0}
```

reasoning:

I'm the receiver. The sender sent me \$0, so I received \$0. I can only return between \$0 and what I received, which is \$0. So I must return \$0.

The game continues with complete non-cooperation. Everyone is just keeping their endowment.

usage:

```
{
  "input_tokens": 1022,
  "output_tokens": 92,
  "reasoning_tokens": 61
}
```

Message 6 role=user

content:

Round 9

You are the SENDER this round. Last round you were the RECEIVER. The sender sent \$0, that was 0.0% of their total endowment. You received \$0. You returned \$0. So your payoff was 0. Your total earnings across all rounds are \$16.89. How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

The game has been in complete defection mode for multiple rounds now. No trust, no cooperation. I'll continue with the Nash equilibrium strategy.

```
{'send': 0}
```

reasoning:

We're now at Round 9. The pattern has been:

- Complete breakdown of cooperation
- No one has sent anything for the last several rounds (Rounds 4-8)

My total earnings are \$16.89 across all rounds. Let me think about the history:

- Early rounds: I tried sending money but got poor returns
- Recent rounds (4-8): Everyone sends \$0

There's absolutely no incentive to send anything now. The equilibrium is mutual defection, and breaking from it would just mean I lose money.

I'll continue with \$0.

usage:

```
{
  "input_tokens": 881,
  "output_tokens": 173,
  "reasoning_tokens": 132
}
```

Message 8 role=user**content:**

Round 10

You are the RECEIVER this round. Last round you were the SENDER. You sent \$0, that was 0.0% of your total endowment. This was tripled to \$0.

the receiver returned \$0. So your payoff was 5.

Your total earnings across all rounds are \$21.89.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

I received \$0, so there's nothing to return.

```
{'return': 0}
```

Reasoning:

I received \$0 again, so there's nothing to return. The game continues with complete defection

- no one is sending anything.

Total earnings: \$21.89

Received this round: \$0

Can return: \$0 (since I received nothing)

Response should be:

```
{'return': 0}
```

Usage:

```
{
  "input_tokens": 880,
  "output_tokens": 99,
  "reasoning_tokens": 68
}
```